

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 15: Automatic Monitor-Based Microwave Photonic Systems



Sam Palermo
Analog & Mixed-Signal Center
Texas A&M University

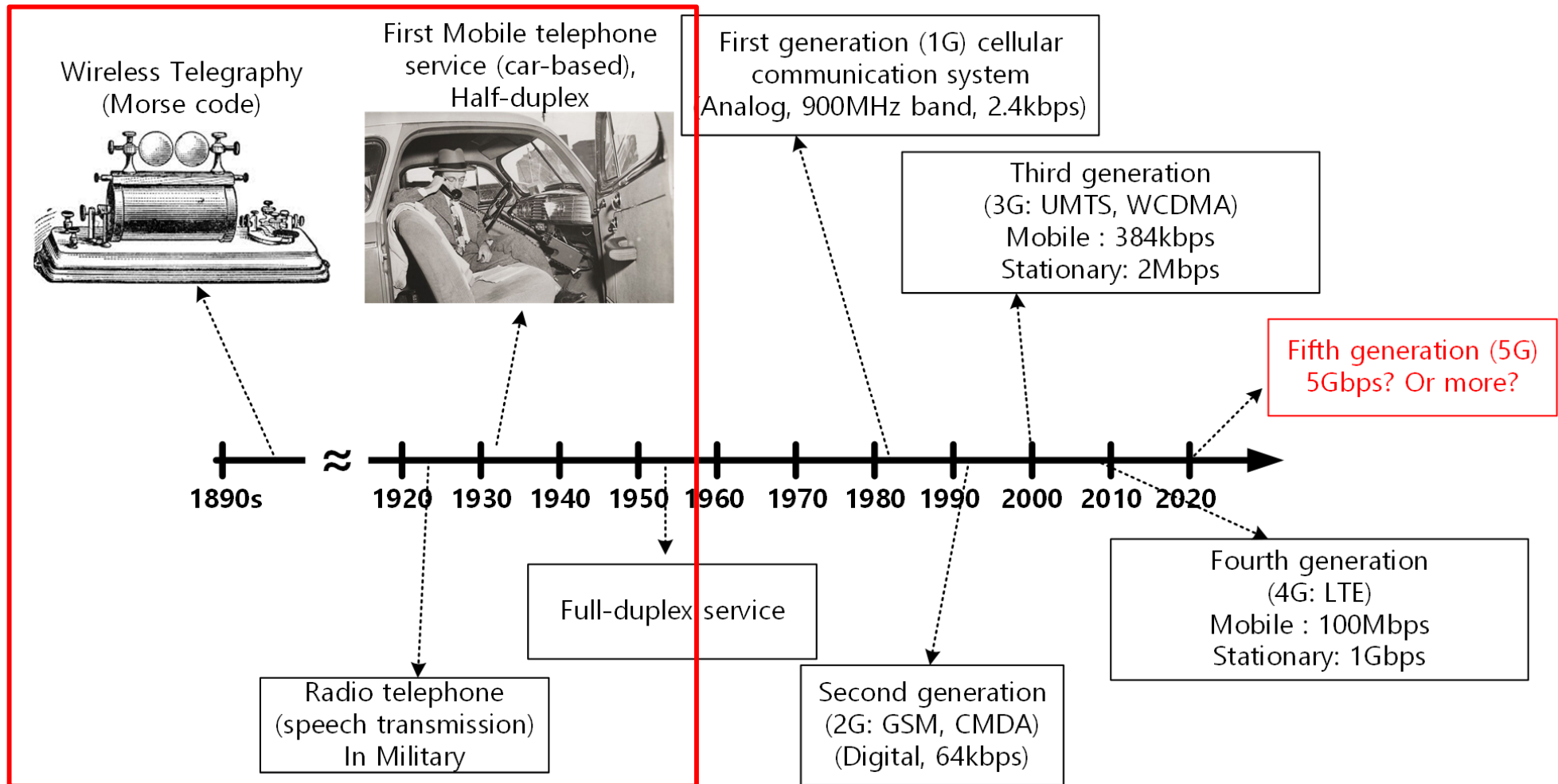
Announcements

- Project Final Report Due Apr 28
- Exam 2 May 4 (1PM-3PM)
 - In class
 - One double-sided 8.5x11 notes page allowed
 - Bring your calculator
 - Covers through Lecture 13

Outline

- Motivation
- Monitor-Based Tuning Principles
- Automatic Filter Tuning
- Automatic Optical Beamforming Network Tuning
- Conclusion

Development of Wireless Communication

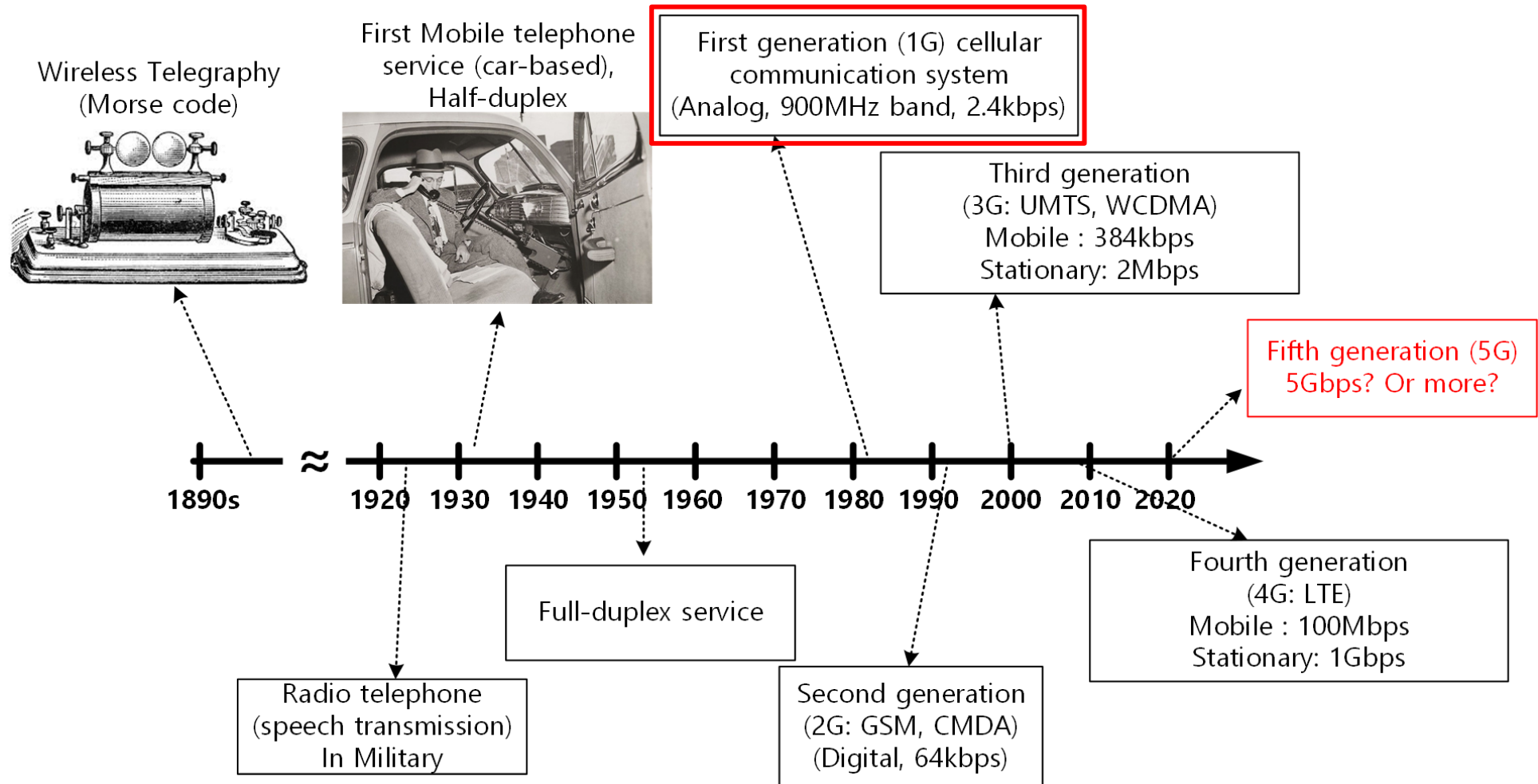


First Wireless communication (1890s) : Telegraphy

First speech transmission (1920s) : Military between naval ships

First Mobile telephone service (1930s) : Taxi <-> Telephone exchange office

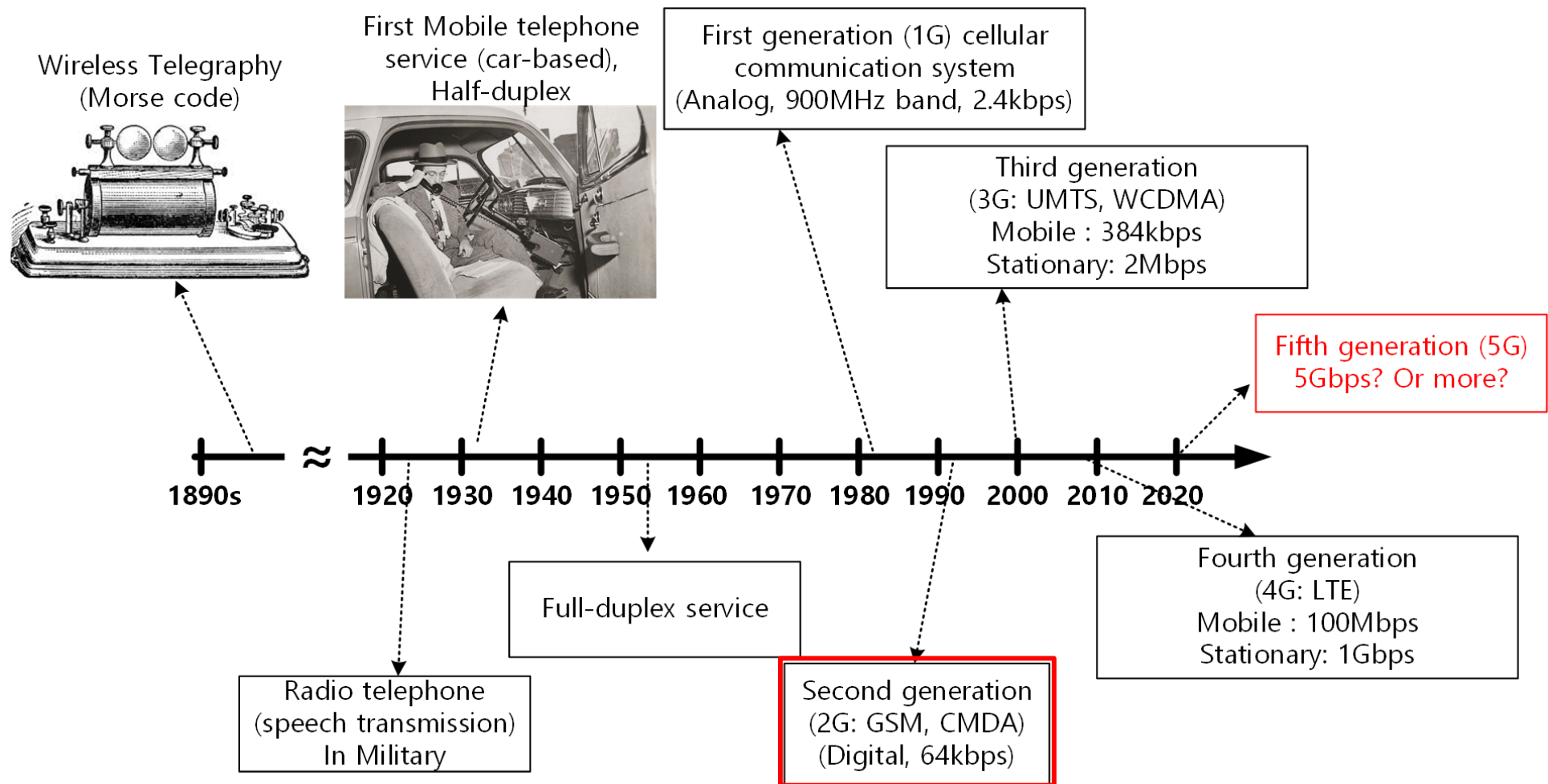
Wireless Communication (1G)



First cellular communication system (1G) : Analog signal, Analog systems

- (-) Poor voice quality, Poor battery life, Large phone size, Limited Capacity

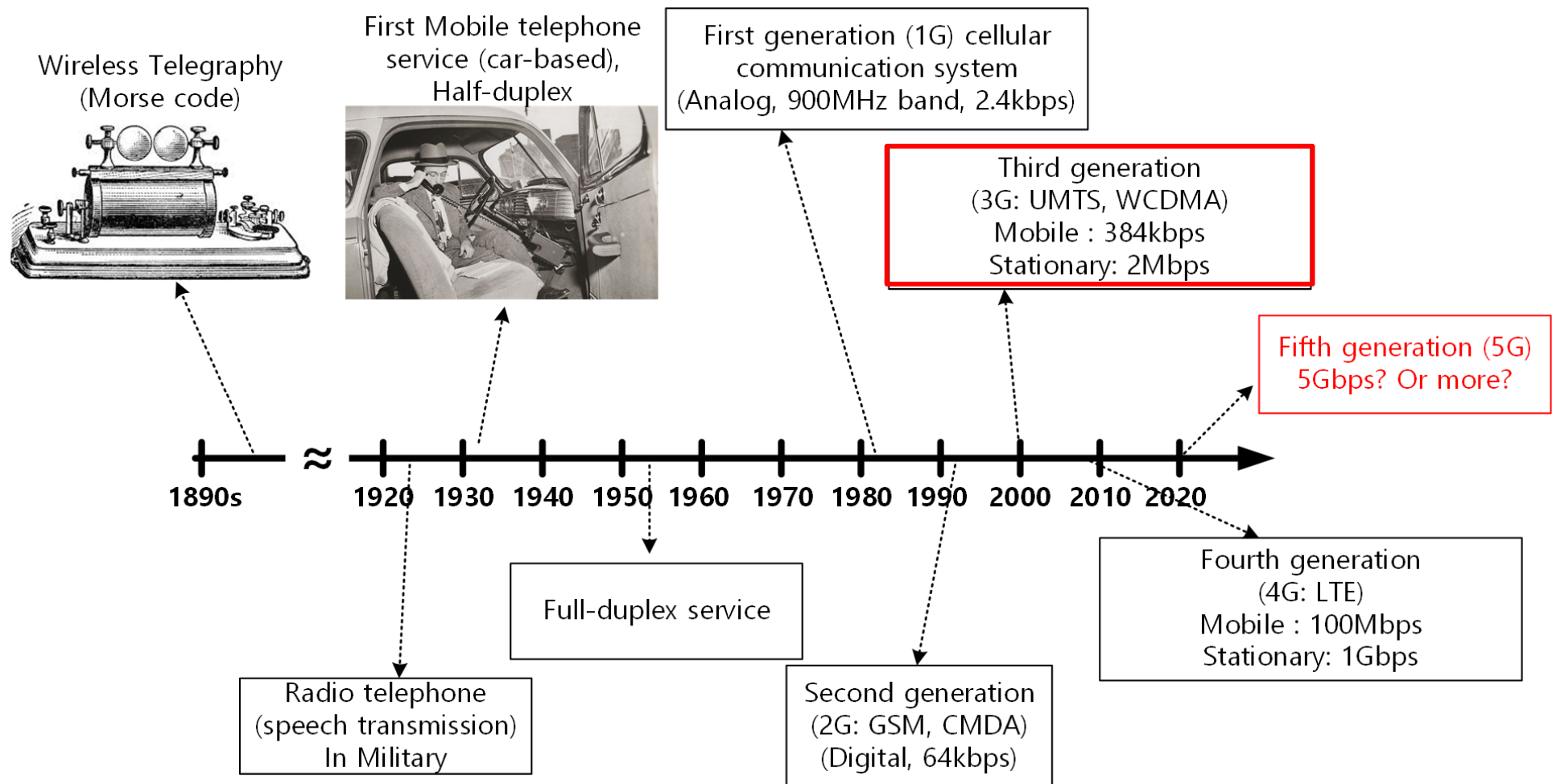
Wireless Communication (2G)



Second cellular communication system (2G) : Digital signal, Digital systems

- Digital Systems, Ability to send SMS, Voice encrypted
- (-) Low data-rate (64kbps)

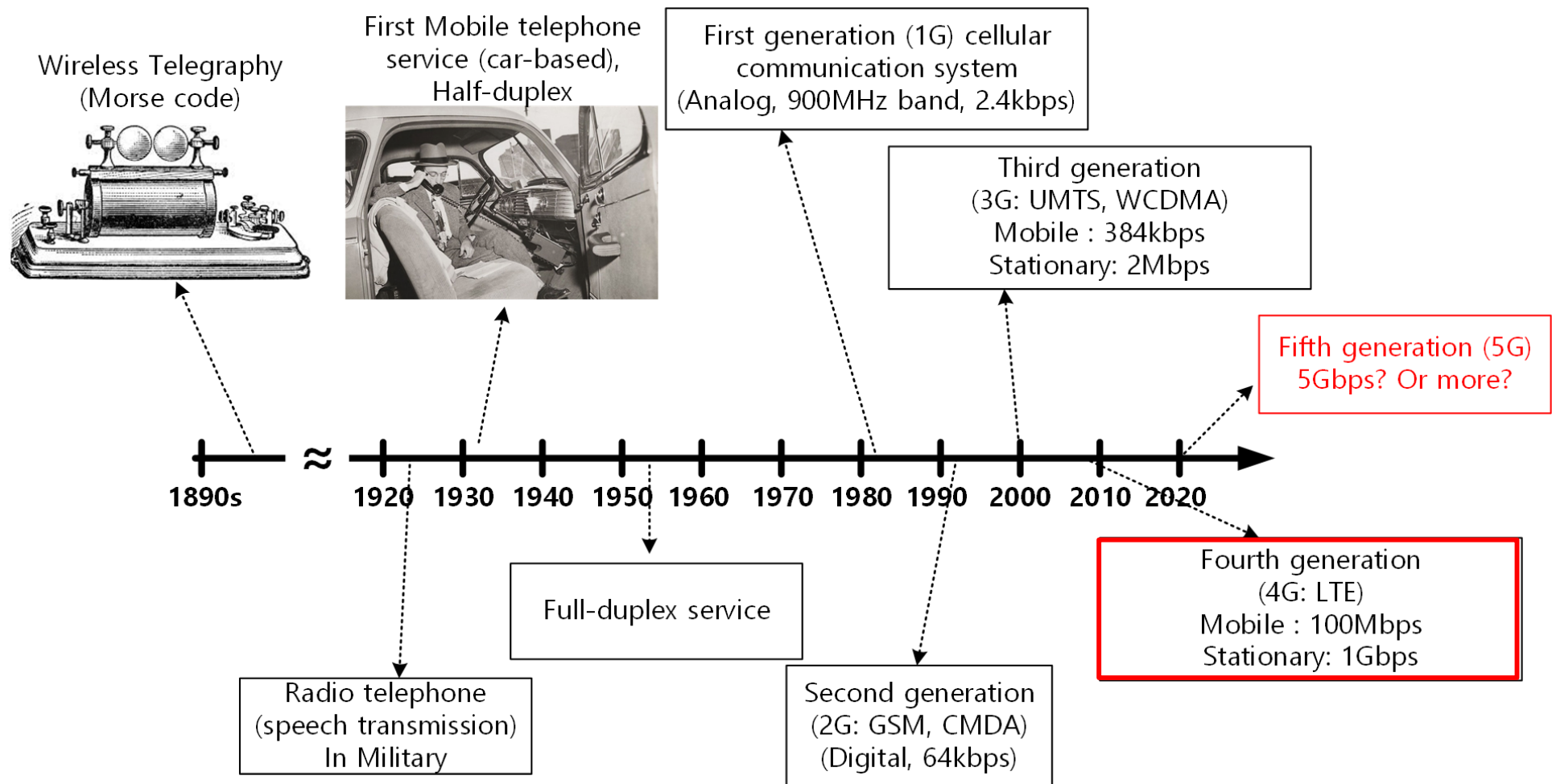
Wireless Communication (3G)



Third cellular communication system (3G) : Large capacities, Broadband

- Send/Receive large email messages, Internet access, TV streaming

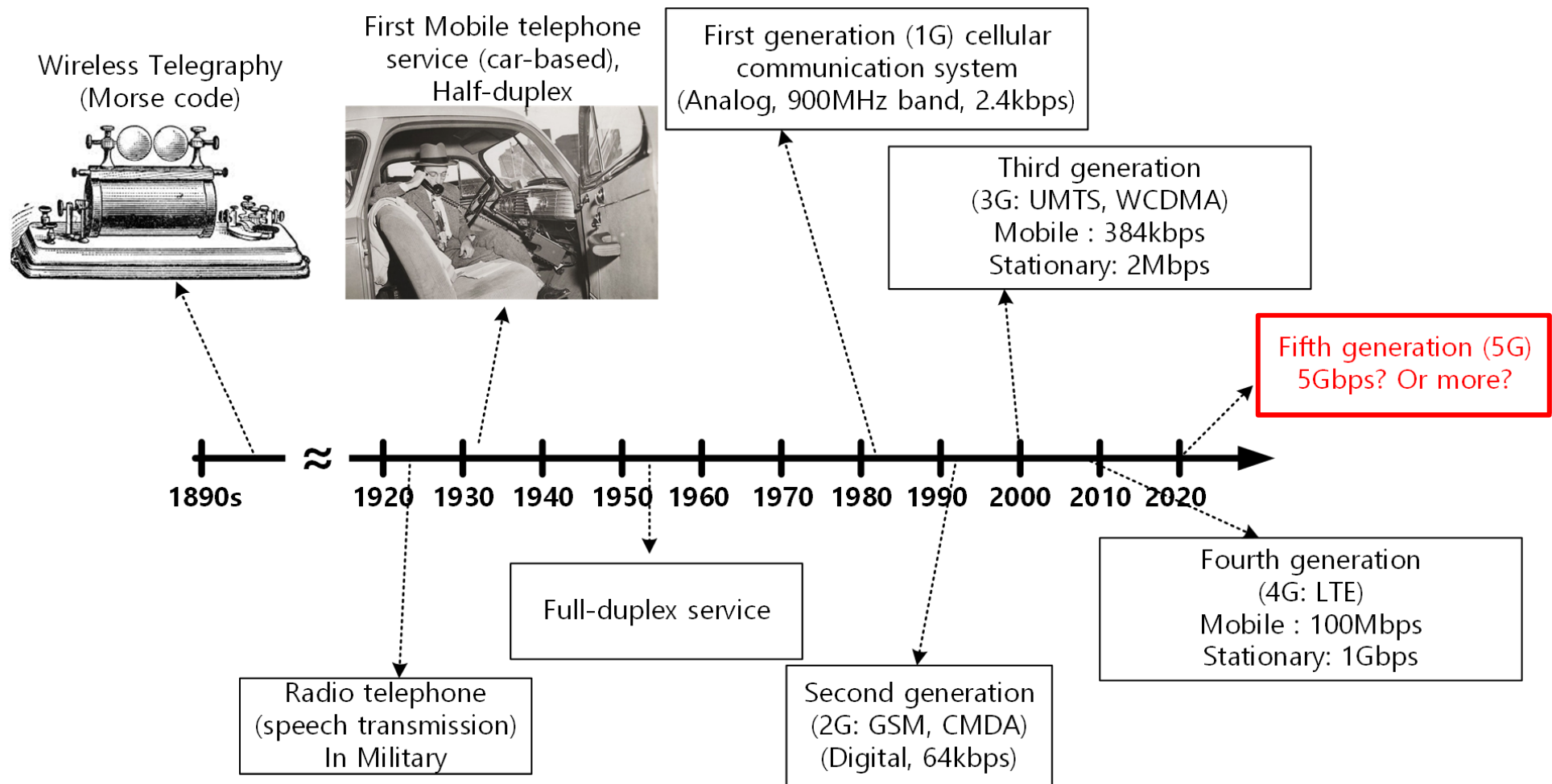
Wireless Communication (4G)



Fourth cellular communication system (4G)

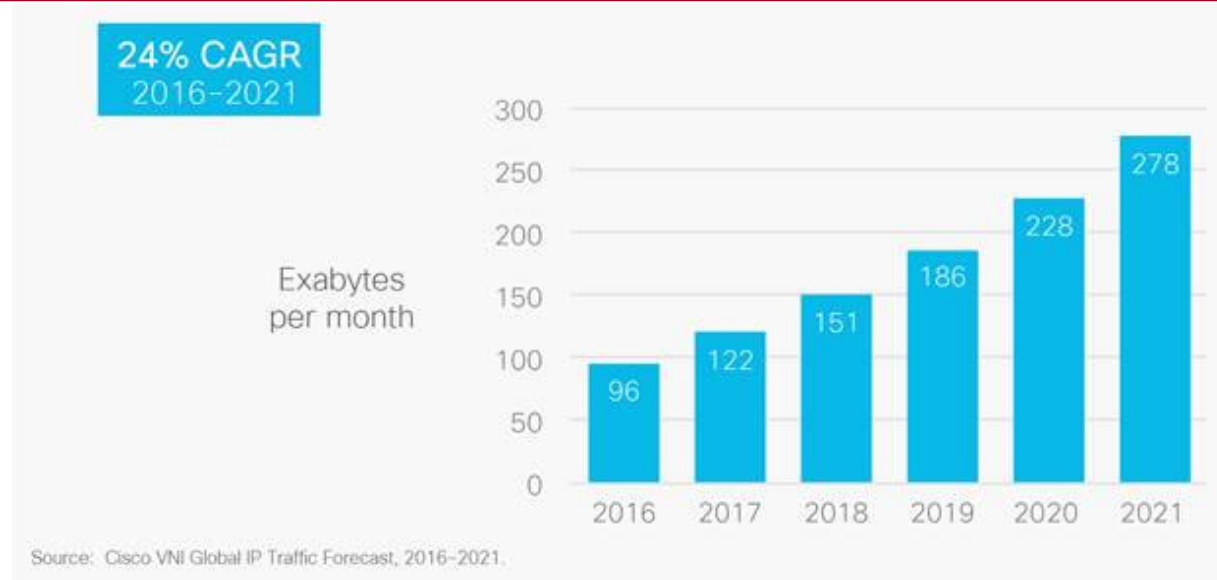
- Higher data rates and expanded multimedia services

Wireless Communication (5G)



Fifth cellular communication system (5G)

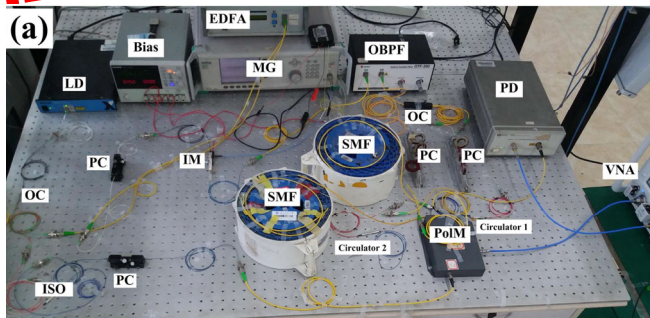
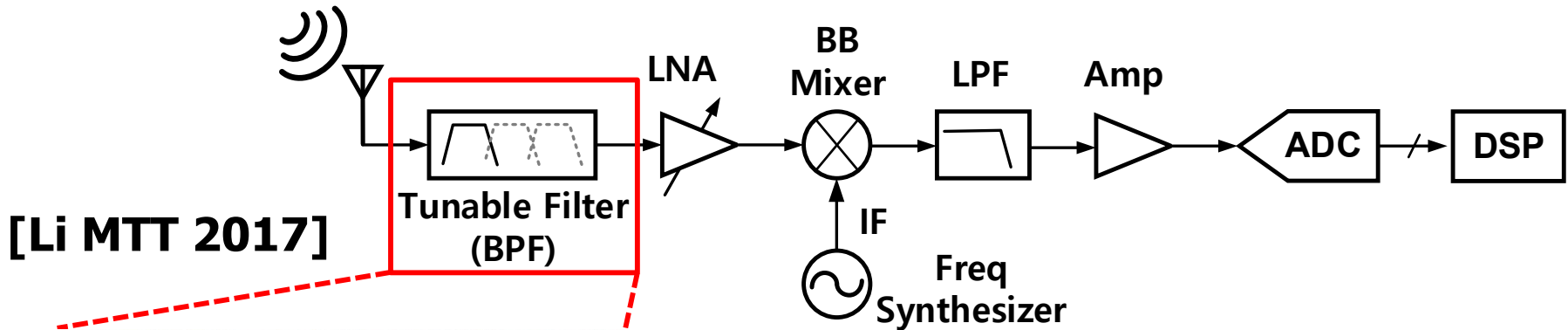
Development of Wireless Communication



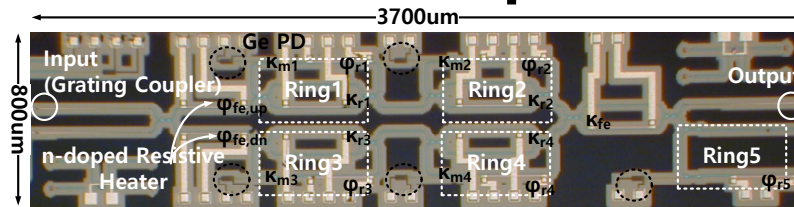
[Cisco Visual Networking Index ,2016]

- Continuously Increasing data traffic
- Fifth cellular communication system (5G)
 - Still no consensus on frequency bands, architecture of network.
 - To meet future data traffic technology evolution is required
 - Reducing the cell size (higher Integration)
 - Building massive multiple-input/output system (MIMO)
 - Shifting to higher frequency bands for wider amount of spectrum
- Challenging to meet with existing electrical systems

Microwave Photonics



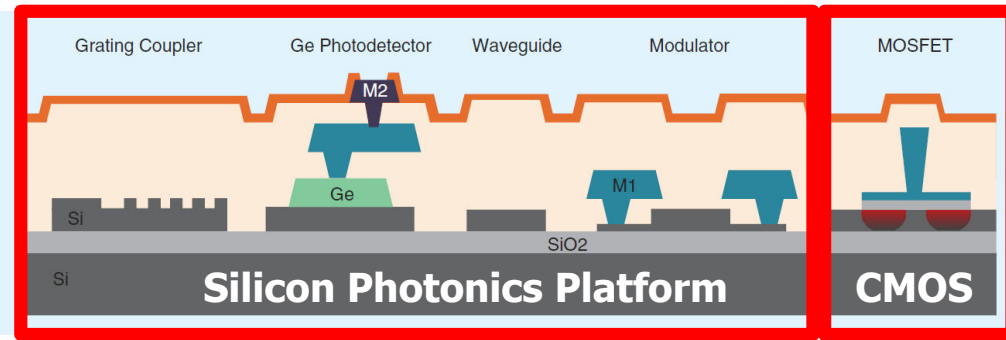
Silicon Photonic Implementation



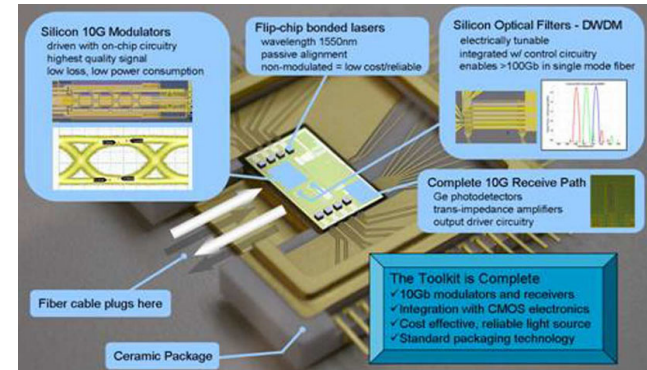
[Choo JLT 2018]

- Advantages
 - Operating at optical frequencies offers extremely high bandwidth
 - Low-loss transmission is possible over optical fibers
 - Orders of magnitude improvement in frequency tuning
- Traditionally systems implemented with bulky discrete components
- Silicon photonic implementations offers significant size, weight, and power improvements

Silicon Photonics



[M. Hochberg, *IEEE Solid-State Circuits Mag*, 2013]



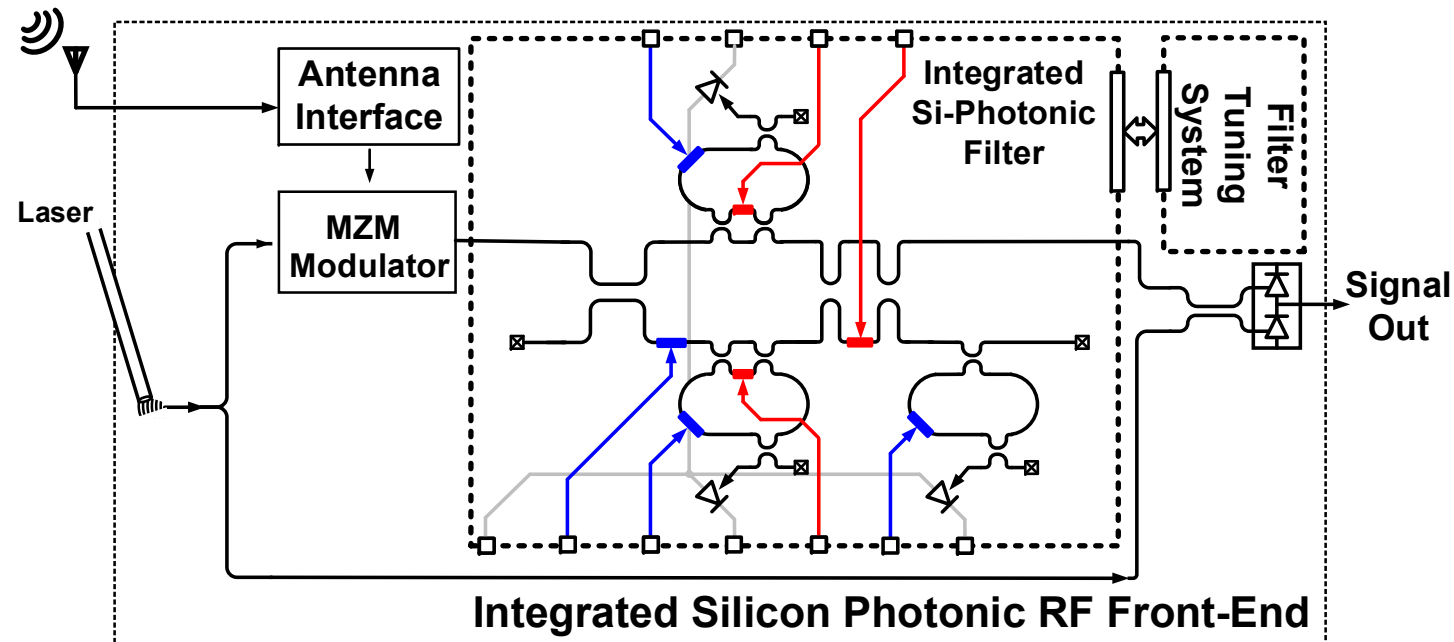
[C. Gunn, *IEEE Micro*, 2006]

- Motivation for **Silicon** Photonics^[1] (SiP)
 - Availability of high-quality SOI wafer
 - High index contrast between Silicon and SiO₂ offers strong optical confinement
 - Ideal platform for planar waveguide circuits
- Compatibility with the mature silicon IC manufacturing
 - Reuse of mature CMOS fabrication infrastructure
 - Monolithic integration with CMOS chips
- Emergence of Silicon photonic integrated circuits made RF photonics promising candidates for the future wireless communication systems^[2]

[1] A. Safarian, RFIC symposium, 2007

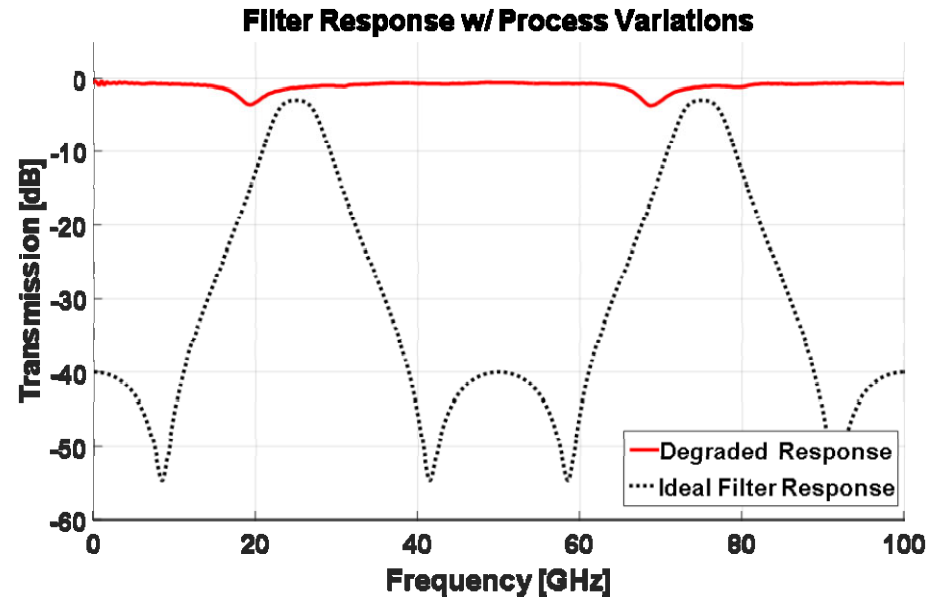
[2] Y. Wu, IEEE TCAS II, 2003

Silicon Photonic mm-Wave Front-End



- Silicon photonic platforms offer the ability to integrate many photonic circuits on a single die
- Micro/mm-wave photonic filters are promising candidate for future wideband receivers since they can support wide bandwidth and dynamic filtering over a broad spectral range

Si-Photonic Filters w/ Process Variations

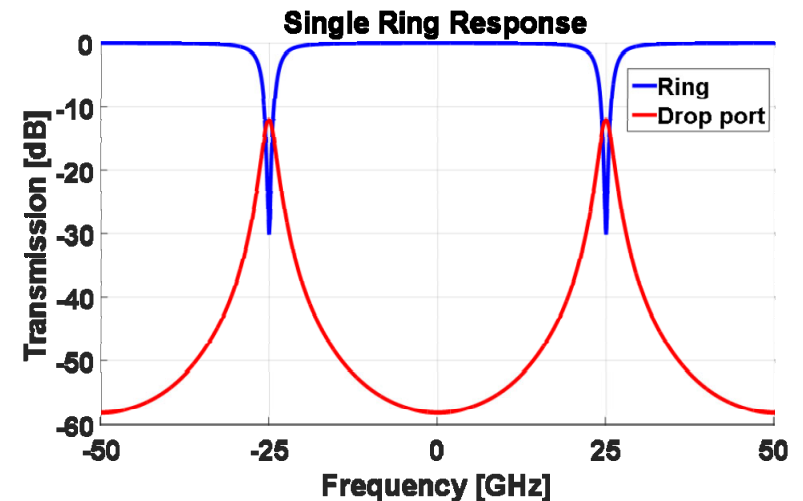
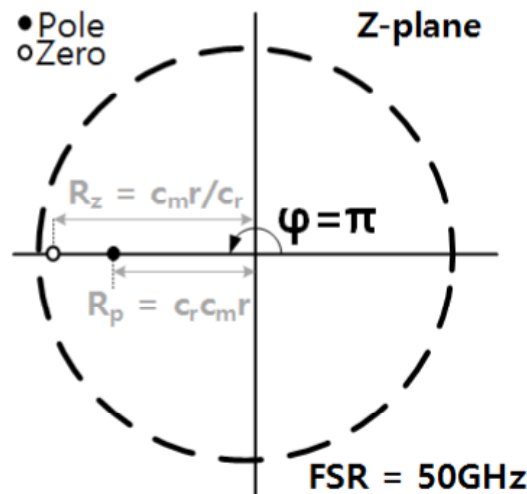
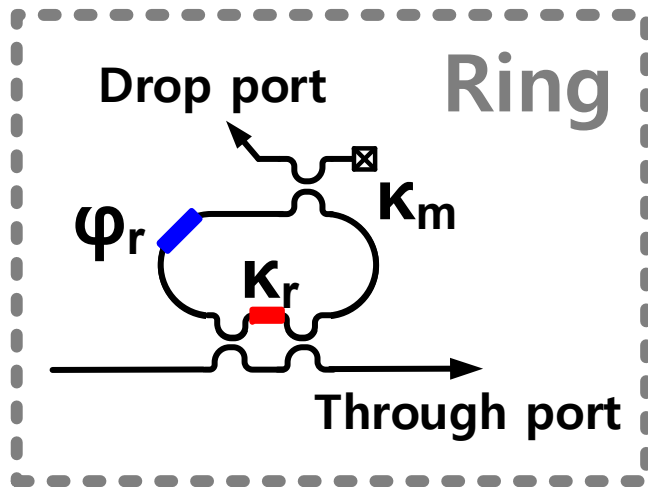


- Photonic devices are sensitive to process and temperature variations
 - Filter responses degrade significantly due to process variations
 - Center frequency shifts with temperature variations
- Manual calibration with spectrum analyzer is expensive, time consuming, and prone to human errors
- Need precise automatic calibration solution

Outline

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- Monitor-Based Tuning Principles
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Ring Resonator Response



$$H_{through}(z) = \frac{c_r - c_m r e^{j\varphi_r} z^{-1}}{1 - c_m c_r r e^{j\varphi_r} z^{-1}},$$

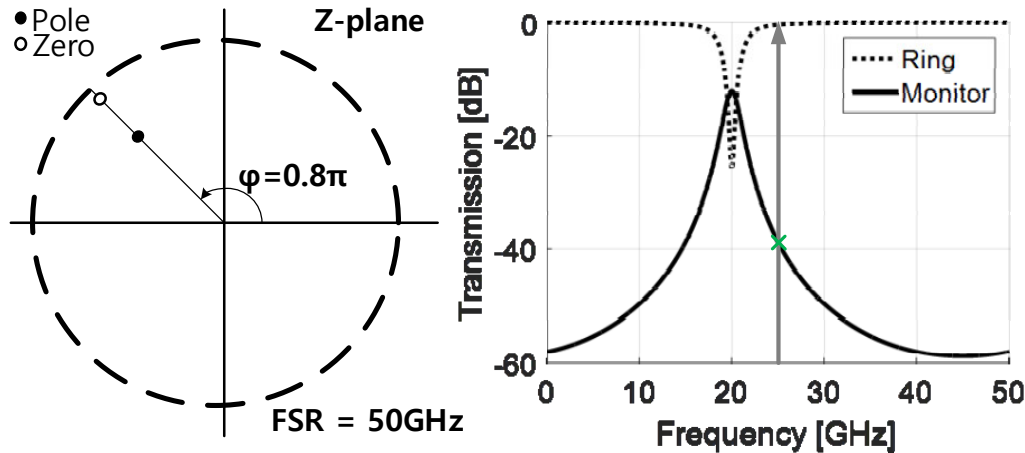
$$H_{drop}(z) = \frac{s_m c_r r e^{j\varphi_r} z^{-1}}{1 - c_m c_r r e^{j\varphi_r} z^{-1}},$$

$$\text{where } c = \sqrt{1 - \kappa}, \quad s = \sqrt{\kappa}$$

- Ring's through and drop port responses are complementary with the notches and peaks in alignment

Monitor-Based Tuning (Resonance)

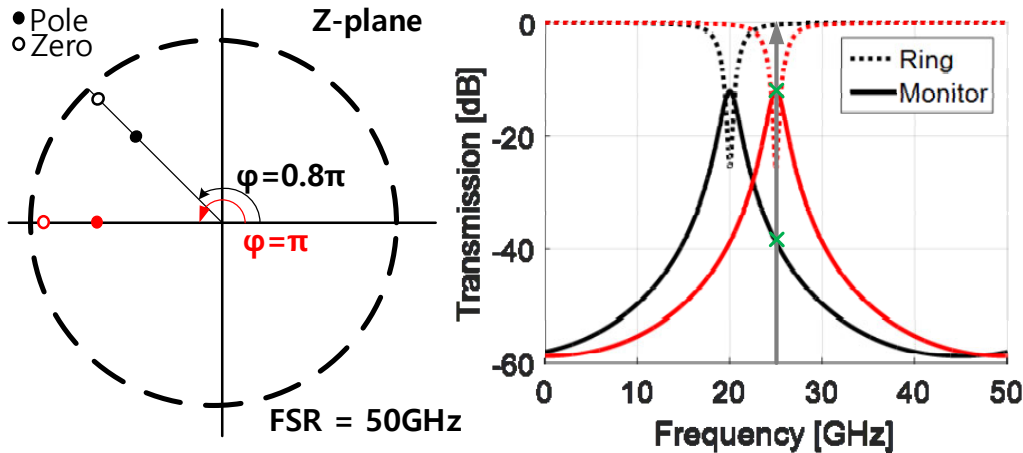
Resonance tuning



- Resonance tuning(φ_r)
 - Ring phase shifter shifts ORR's resonance frequency
 - Resonance is tuned to frequency stimulus by maximizing monitor reading

Monitor-Based Tuning (Resonance)

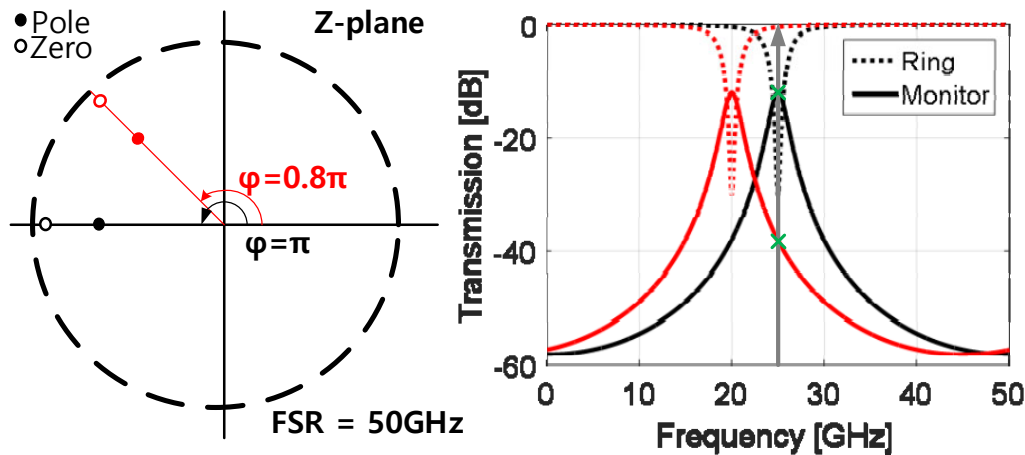
Resonance tuning



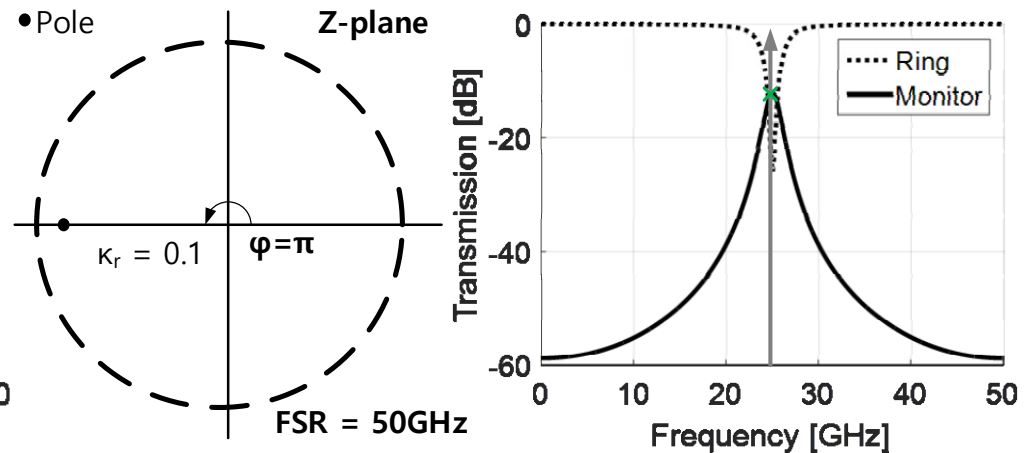
- Resonance tuning(φ_r)
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Monitor-Based Tuning (Coupling)

Resonance tuning



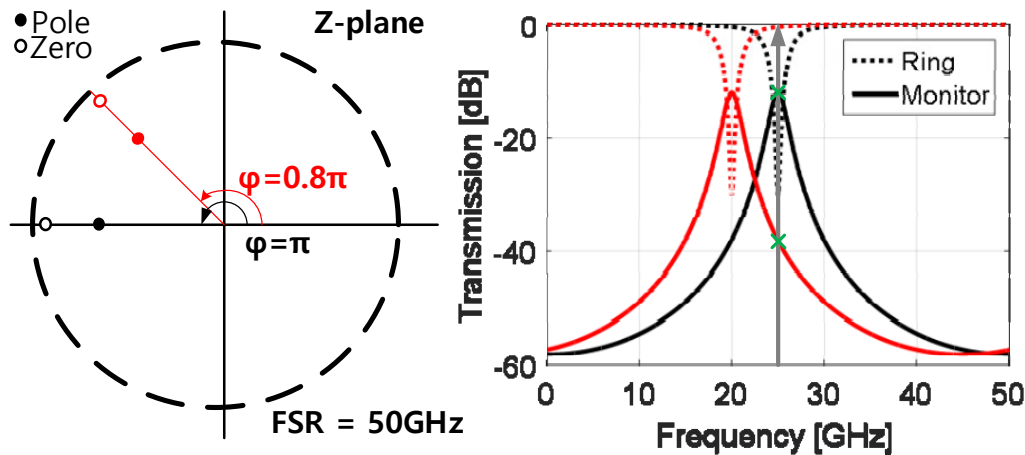
Coupling tuning



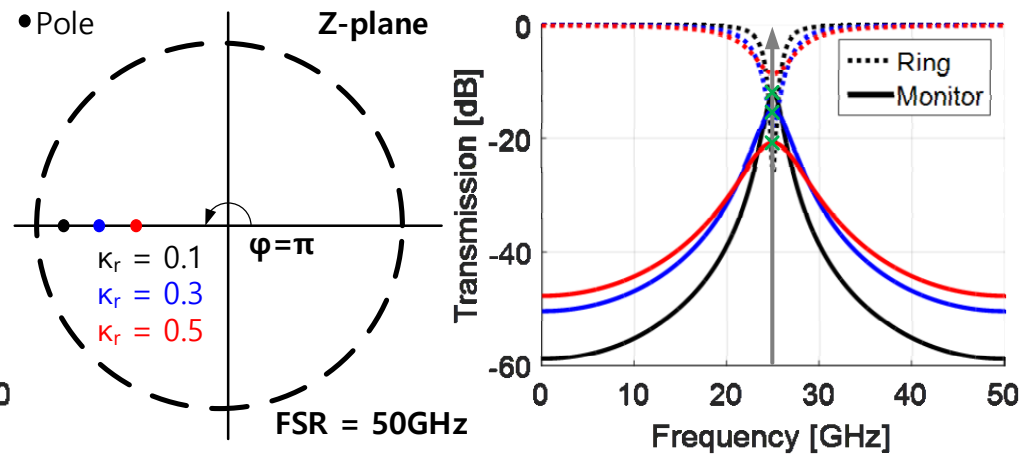
- Resonance tuning(ϕ_r)
 - Ring phase shifter shifts ORR's resonance frequency
 - Resonance is tuned to frequency stimulus by maximizing monitor reading
- Coupling tuning(κ_r)
 - Ring coupler setting changes peak value of the monitor response
 - Monitor response has the maximum reading at critical coupling

Monitor-Based Tuning (Coupling)

Resonance tuning



Coupling tuning



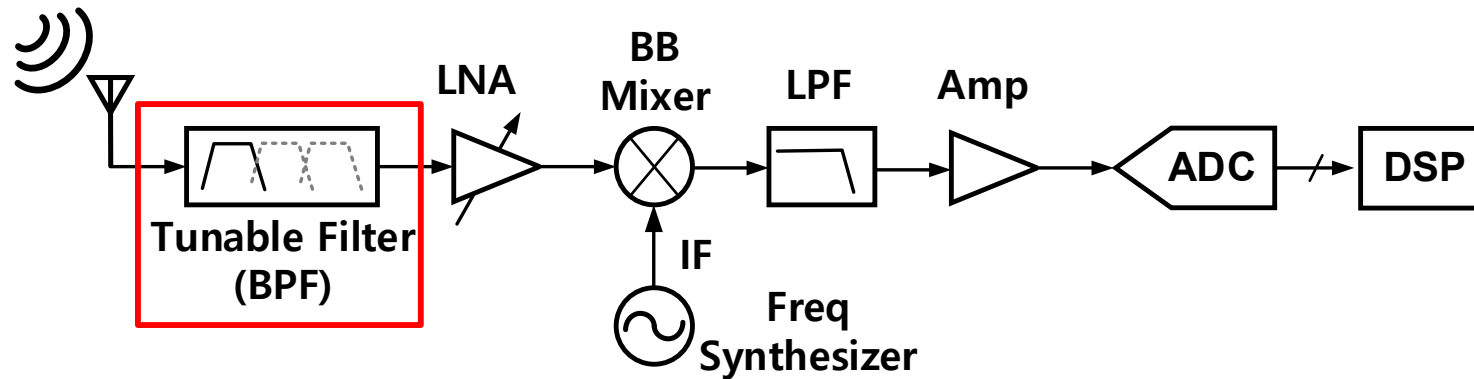
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Broadband mm-wave Receiver

- System block diagram of mm-wave receiver



- Front-end filtering plays the critical role [3]
 - Guaranteeing the RF performance
 - Relaxing subsequent ADC and DSP requirements
- Multi-GHz tuning range, Bandwidth tunability, high out of band rejection are required to fit into the future filter requirements

Limitation of Electrical Filtering Solution

- Off-chip surface acoustic wave (SAW) filters
 - High frequency, multi-band, large tuning range is not feasible^[4]
- Integrated analog filters^[5]
 - On-chip inductor Q-factor limits its selectivity, bandwidth
 - Active nature limit its linearity
- Integrated SAW-less receivers^[6]
 - Proposed for dynamic bandpass filtering
 - Hard to extend operating frequency into mm-wave range



Off-chip SAW filters

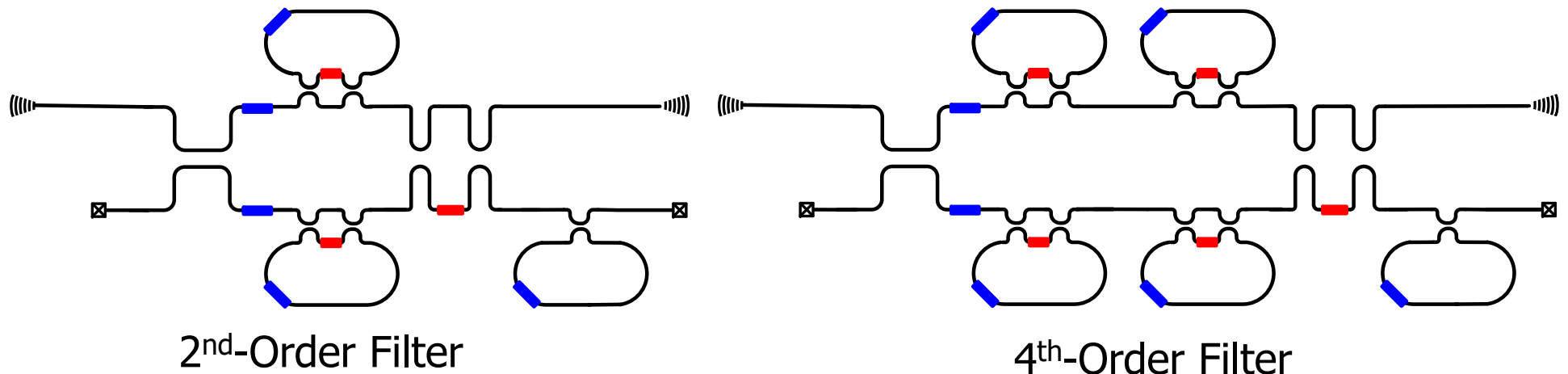
[4] A. Safarian, RFIC symposium, 2007

[5] F. Dulger, IEEE JSSC, 2003

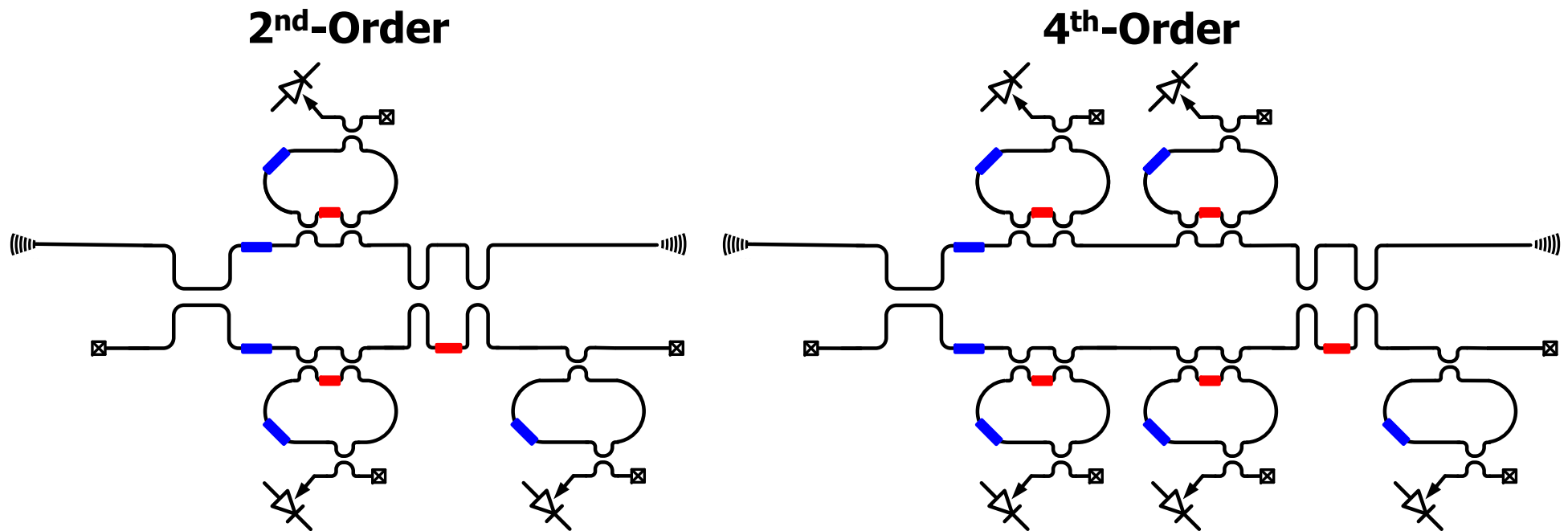
[6] Y. Wu, IEEE TCAS II, 2003

All-pass Based Pole/Zero filter

- Basic pole/zero filter has half rings on top/bottom arms
- MZI couplers are implemented for bandwidth tunability and compensation of fabrication variations
- Additional Ring, end MZI coupler(k_{fe}), and front phase shifters ($\varphi_{fe,up}$, $\varphi_{fe,dn}$) are employed for rejection band tunability

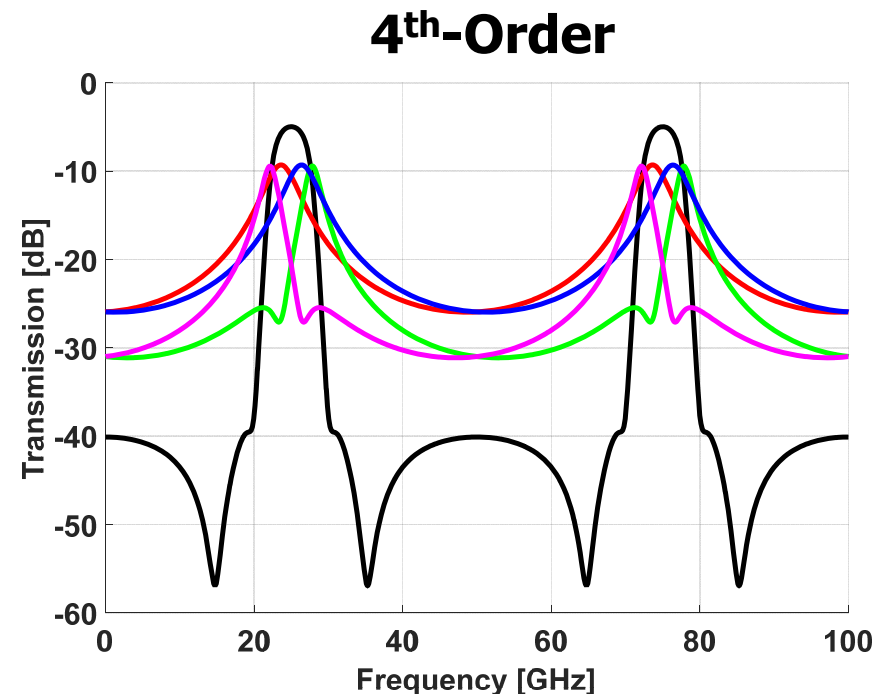
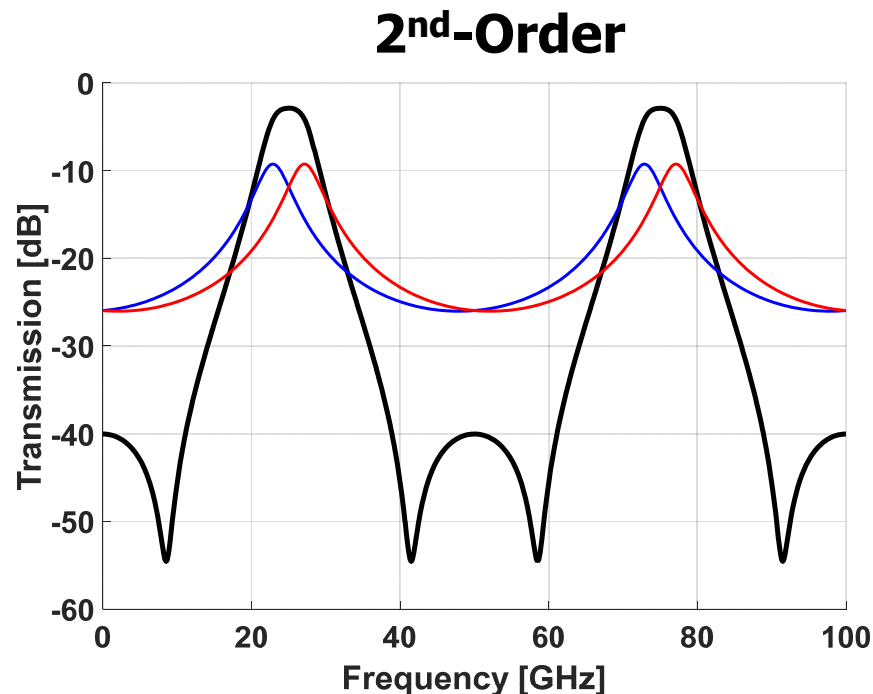


Modified All-Pass-Based Pole/Zero Filter



- Basic pole/zero filter has equal rings on top/bottom arms
- MZI couplers are implemented for bandwidth tunability and compensation of fabrication variations
- Additional ring, end MZI coupler(k_{fe}), and front phase shifters ($\Phi_{fe,up}$, $\Phi_{fe,dn}$) are employed for rejection band tunability
- Drop port with monitor PD are added to each ring to enable monitor-based automatic tuning

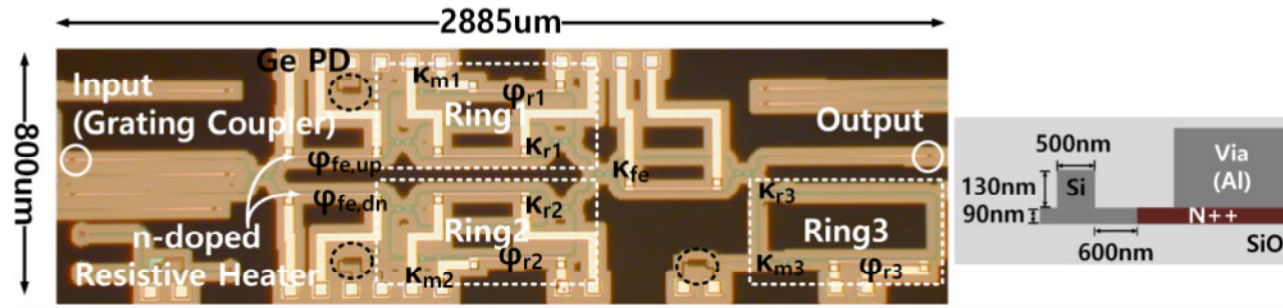
Simulated Filter Responses



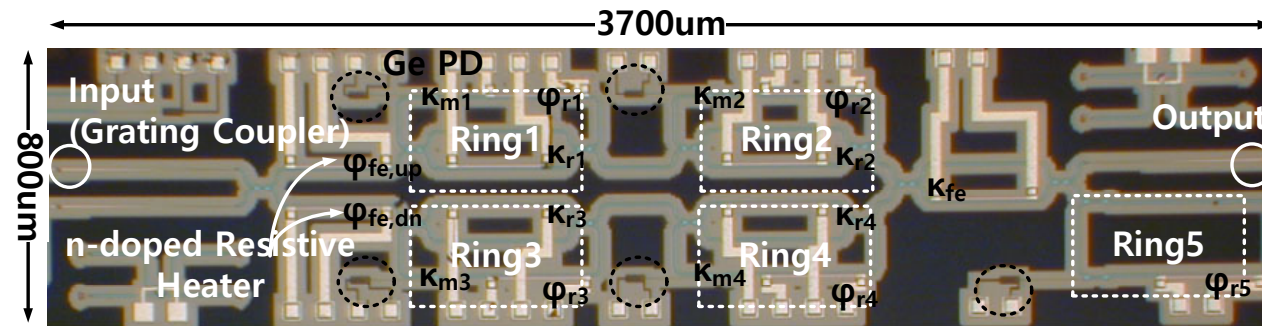
- Centered at 25GHz relative to 1550nm laser wavelength
- Optical waveguide propagation loss (RTL = 0.5dB) produces rounding in the passband
 - 2nd order 3dB bandwidth: 7GHz
 - 4th order 3dB bandwidth: 5GHz
- Monitor responses ($k_m = 0.05$) considered in simulation

Silicon Photonic Optical Filter Prototypes

2nd-Order



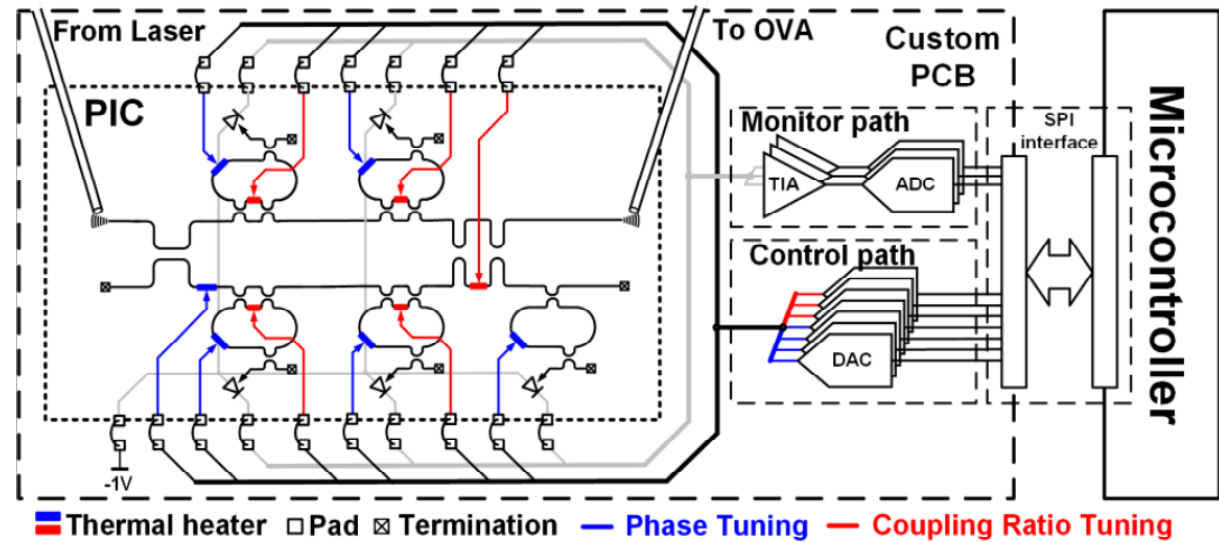
4th-Order



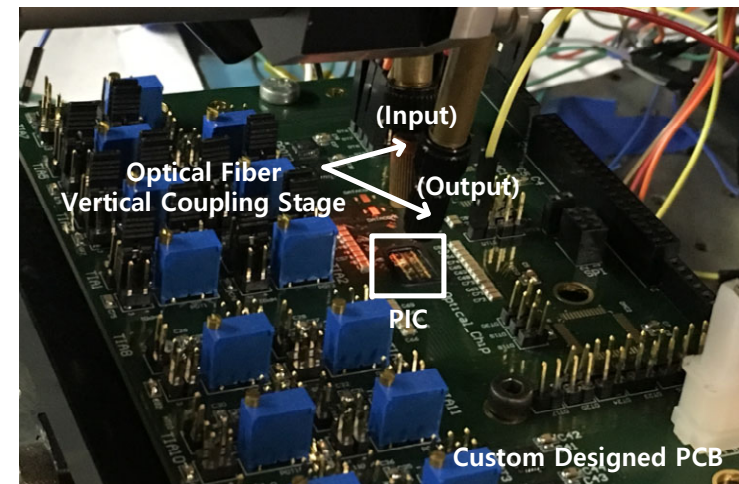
- Fabricated in IME SOI silicon photonics process
- Phase shifters are implemented with resistive heaters
- Rings are designed with 1554 μm circumference to provide a filter response with a 50GHz free spectral range

Filter Tuning System and Procedure

- Automatic tuning system
 - Microcontroller
 - DAC: heater control
 - TIA, ADC: monitor

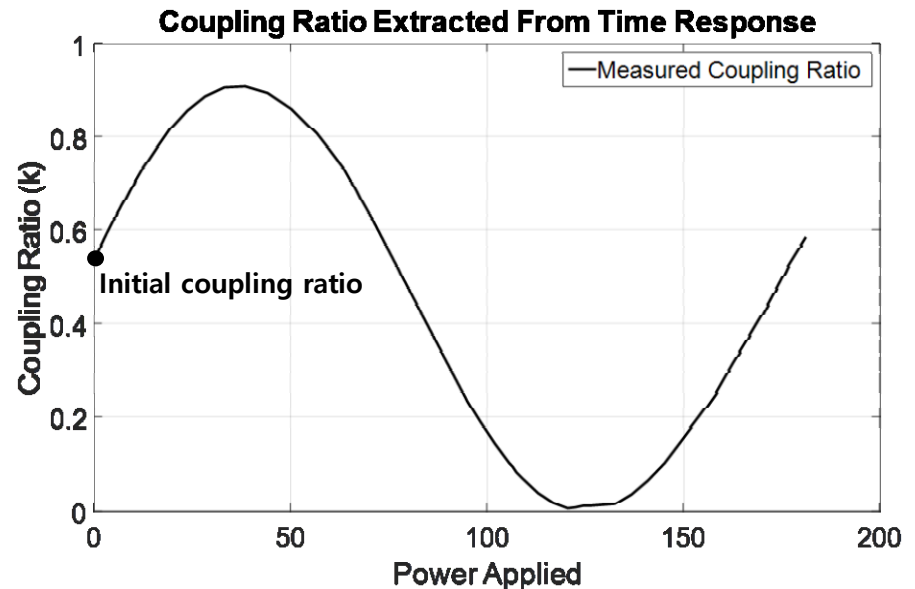
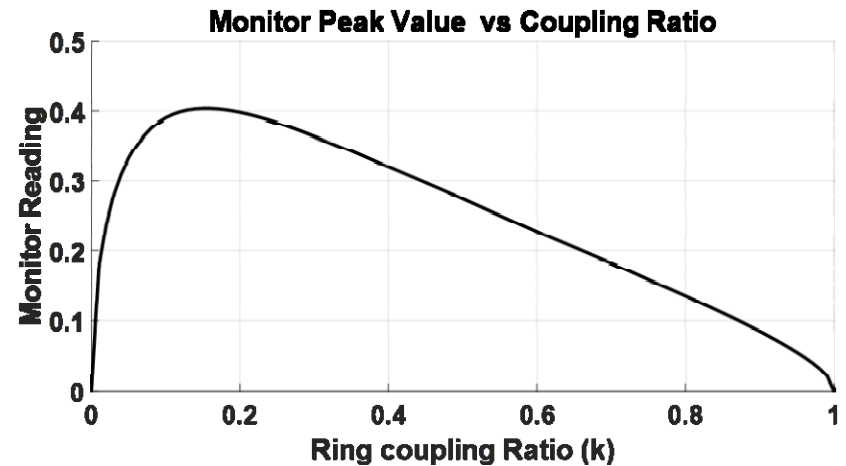


- Tuning procedure
 1. Ring coupler tuning($\kappa_{r1,2}$)
 2. Resonance tuning ($\varphi_{r1,2,3}$)
 3. Rejection band tuning($\varphi_{fe,dn}$, κ_{fe})



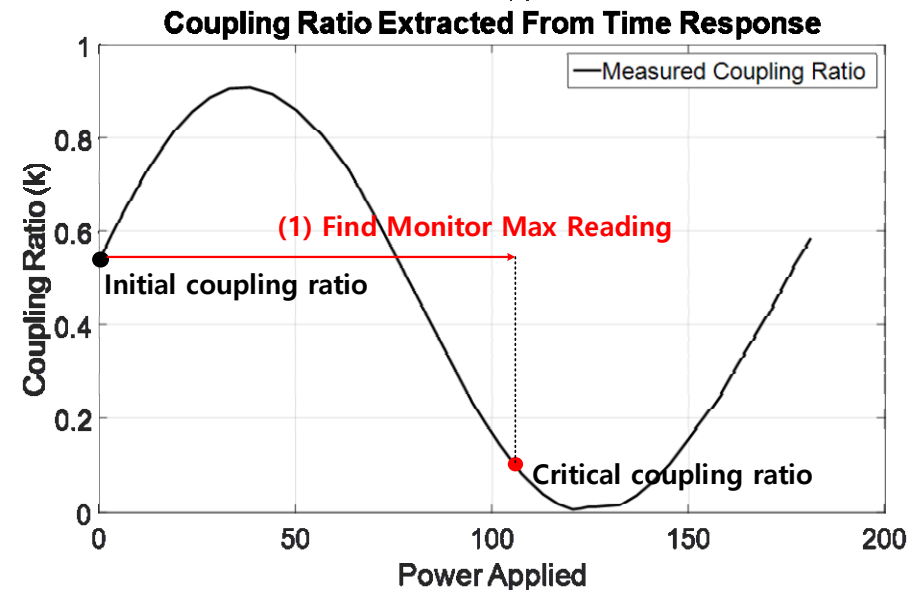
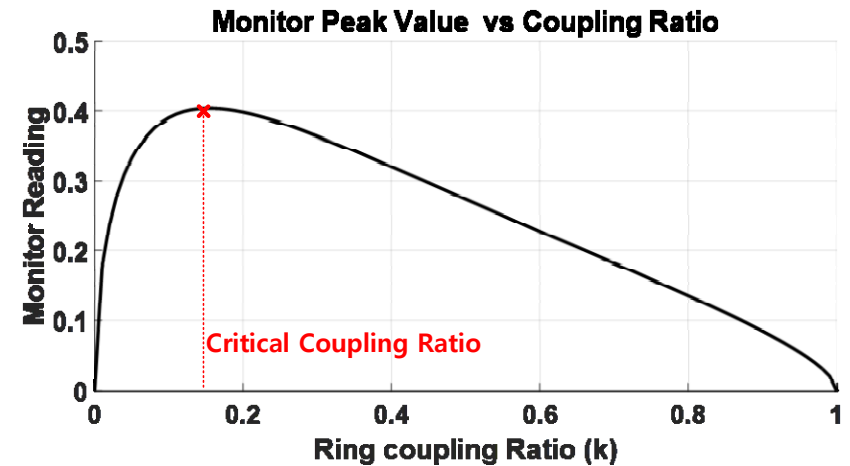
Tuning Procedure (1)

- Step 1: Ring coupler tuning($\kappa_{r1,2}$)
 - (1) Find critical coupling ratio
 - Initial coupling ratio can deviate from the designed value (process variation)



Tuning Procedure (1)

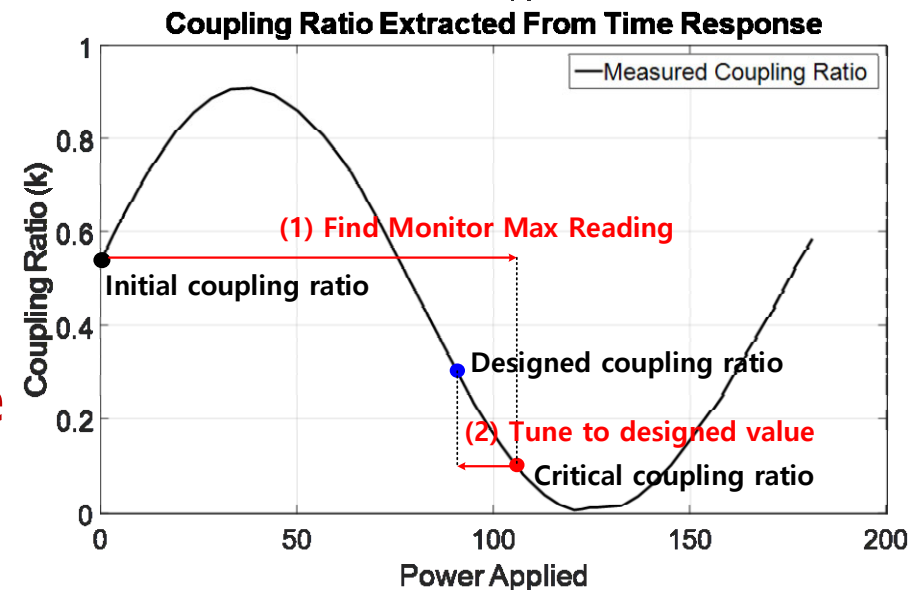
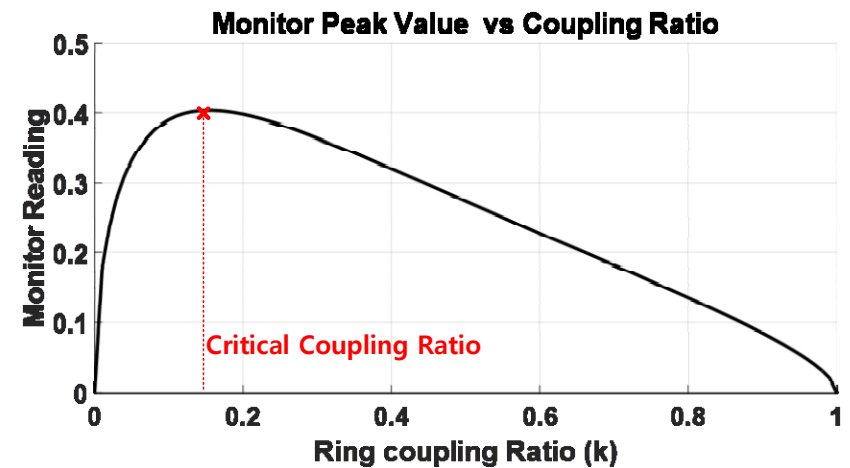
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 - (1) Find critical coupling ratio
 - Initial coupling ratio can deviate from the designed value (process variation)
 - Critical coupling shows maximum monitor reading and serves as a reference point



Tuning Procedure (1)

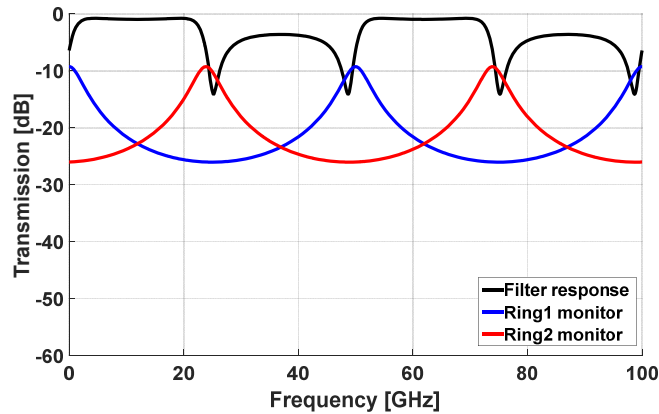
- Step 1: Ring coupler tuning($\kappa_{r1,2}$)
 - (1) Find critical coupling ratio
 - Initial coupling ratio can deviate from the designed value (process variation)
 - Critical coupling shows maximum monitor reading and serves as a reference point
 - (2) Tune to designed coupling ratio
 - Coupling ratio follows MZI characteristic

Coupler tuning also shifts the ring's resonance which is corrected with the ring resonance tuning procedure

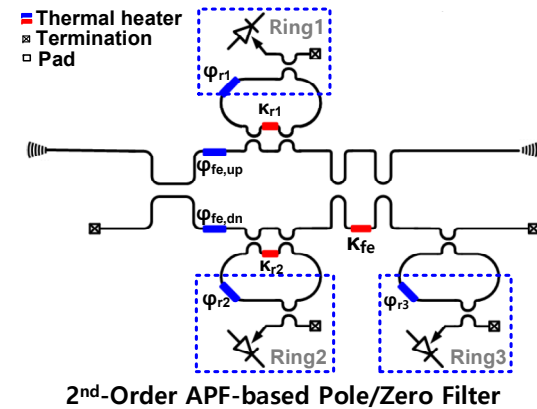


Tuning Procedure (2)

After Step1

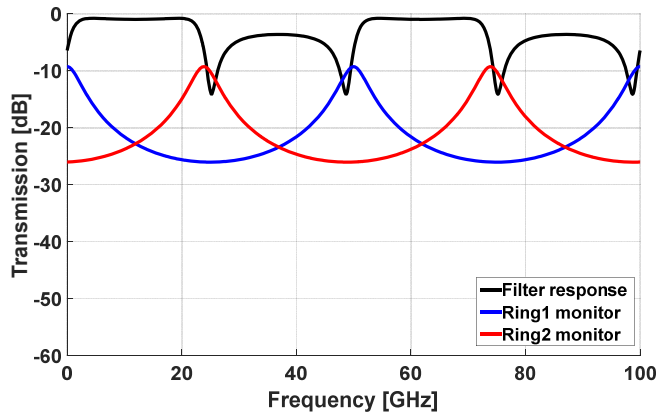


- Step 2: Resonance Tuning($\varphi_{r1,2,3}$)

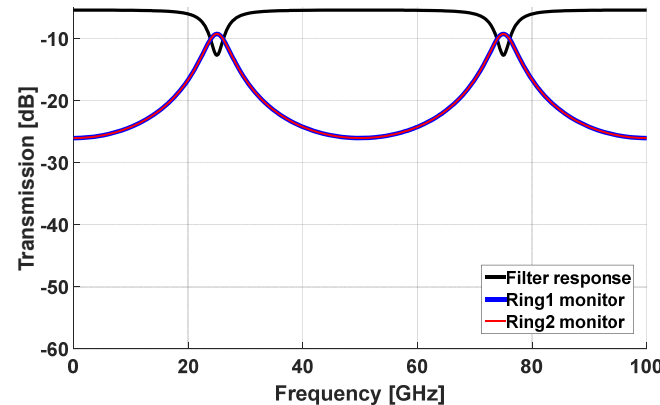


Tuning Procedure (2)

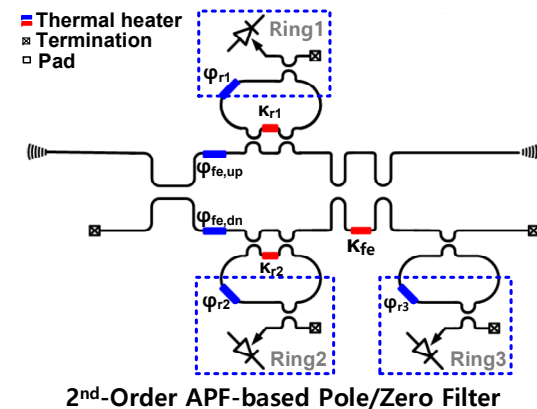
After Step1



Ring1,2 tuned to center frequency

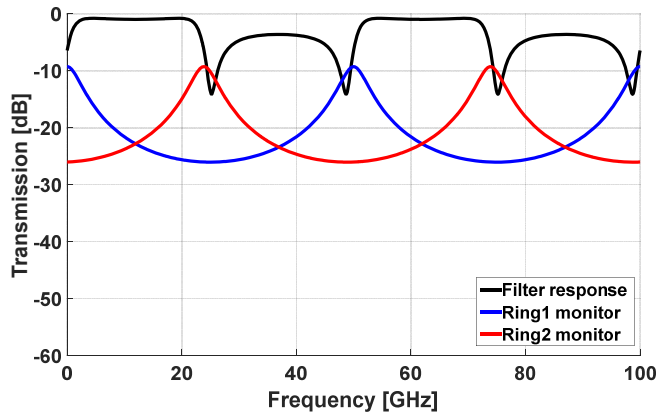


- Step 2: Resonance Tuning($\varphi_{r1,2,3}$)
 - Tune Ring1,2 to the center frequency
 - Ring3 is also tuned to the null frequency for rejection band tuning and reduced sensitivity to thermal cross talk
 - Multiple iterations are performed due to thermal crosstalk

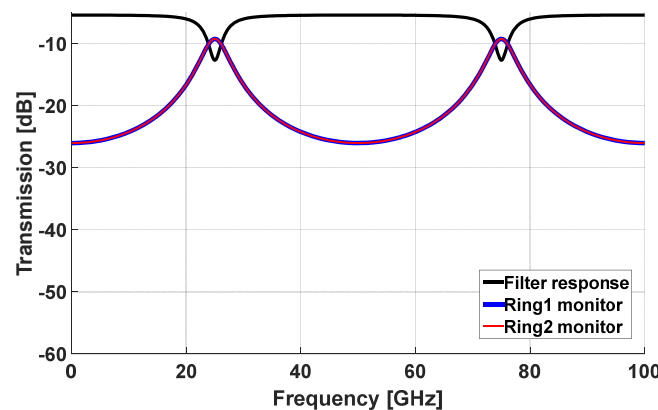


Tuning Procedure (2)

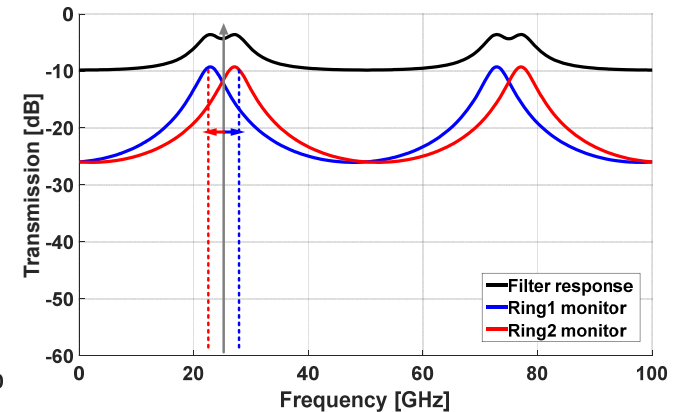
After Step1



Ring1,2 tuned to center frequency

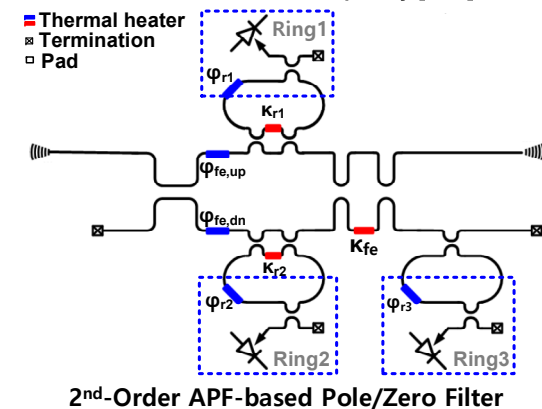


After Step2

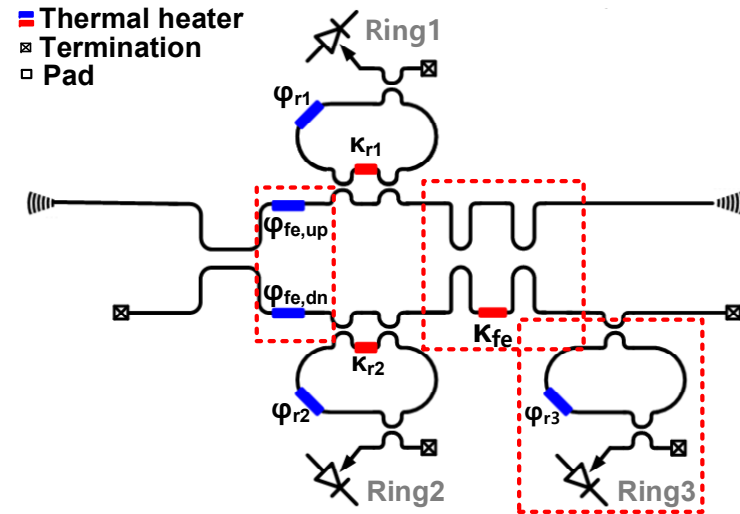
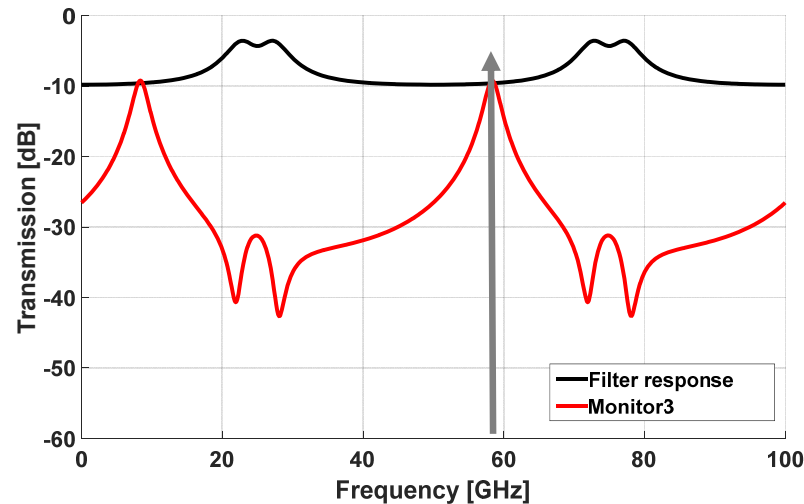


- Step 2: Resonance Tuning($\varphi_{r1,2,3}$)

- Tune Ring1,2 to the center frequency
- Ring3 is also tuned to the null frequency for rejection band tuning and reduced sensitivity to thermal cross talk
- Multiple iterations are performed due to thermal crosstalk
- Ring1,2 are blue/red shifted to yield appropriate monitor reading and set the filter bandwidth

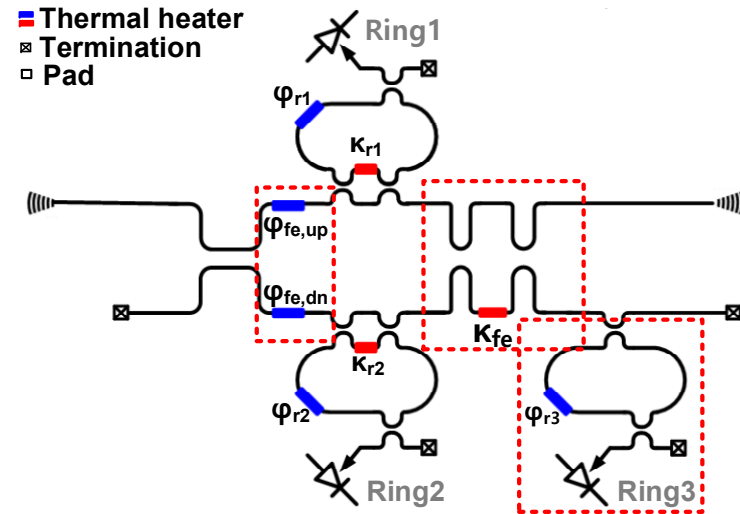
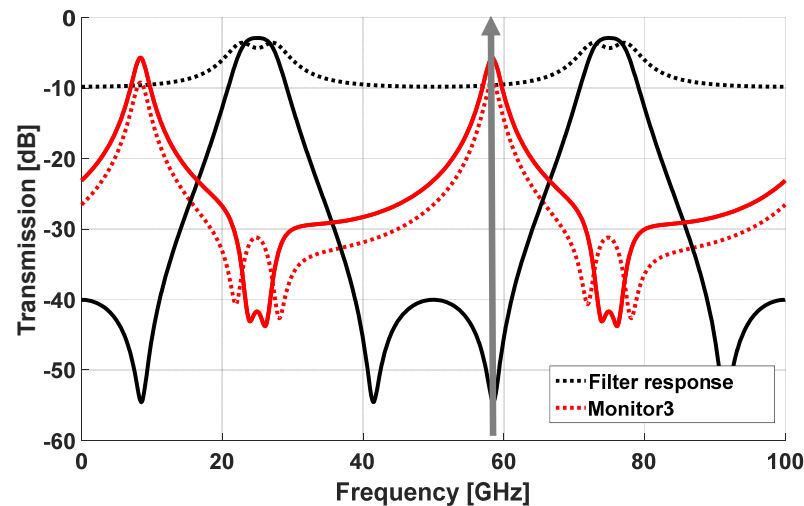


Tuning Procedure (3)



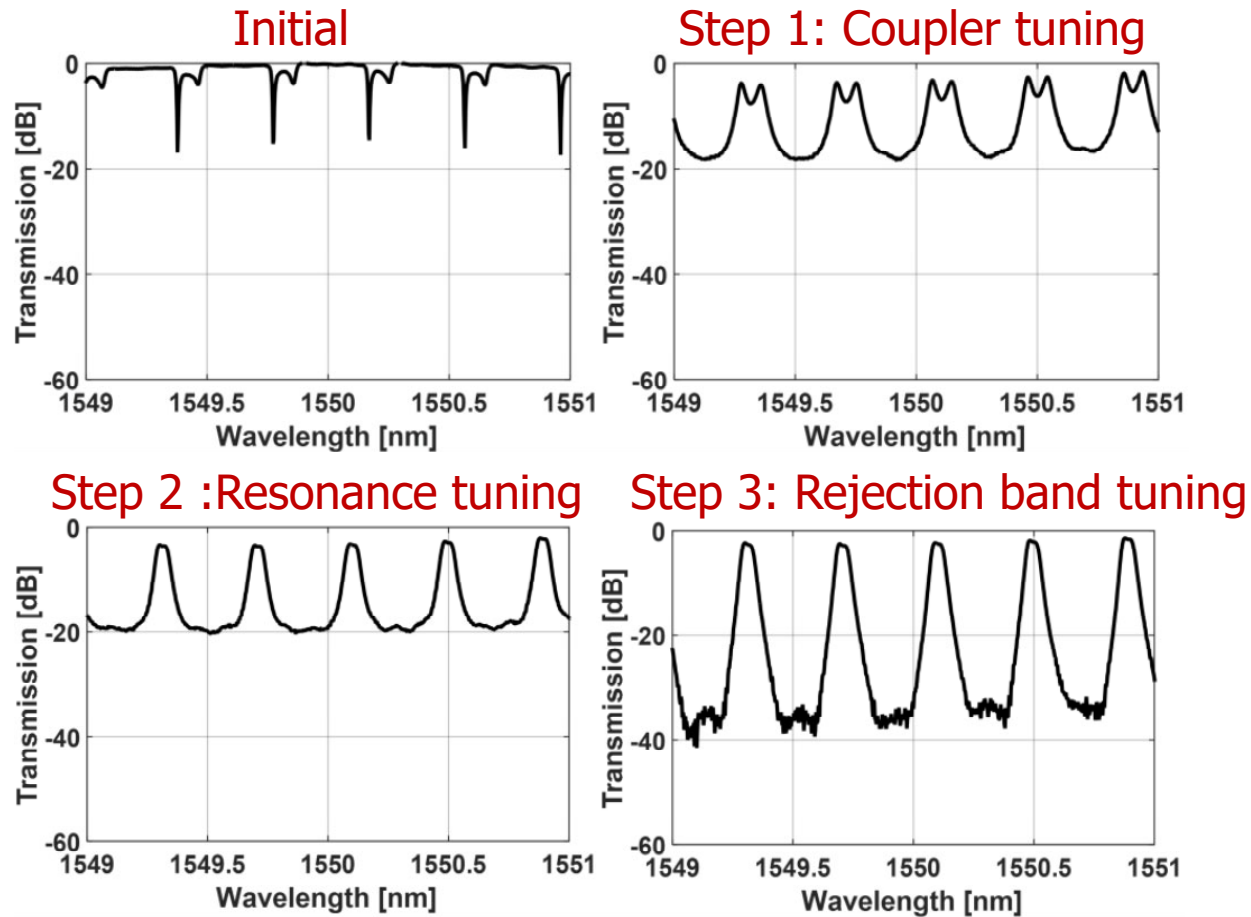
- Step 3: Rejection Band Tuning($\varphi_{fe,dn}$, K_{fe})
 - Ring3 is placed at the complementary port of the filter response
 - Input laser frequency is switched to null frequency of filter response

Tuning Procedure (3)

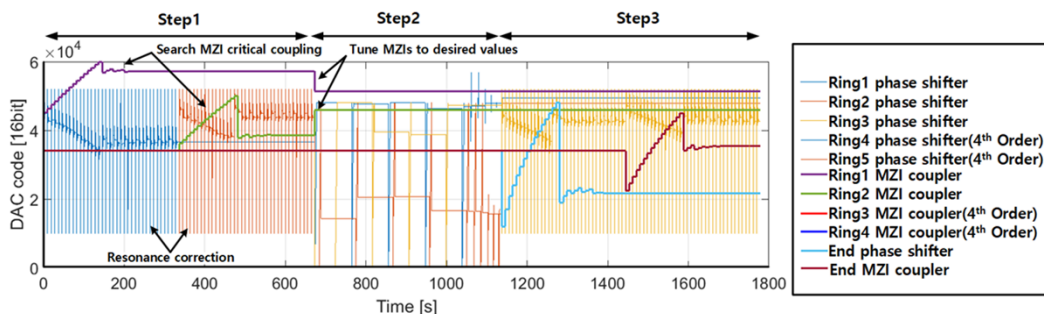


- Step 3: Rejection Band Tuning($\phi_{fe,dn}$, K_{fe})
 - Ring3 is placed at the complementary port of the filter response
 - Input laser frequency is switched to null frequency of filter response
 - Maximizing the monitor3 reading at the null frequency lowers the out of band rejection of the filter response
 - While tuning rejection band, resonance tuning of ring3 is also performed to monitor the maximum value

Measured 2nd-order Filter Initial Calibration

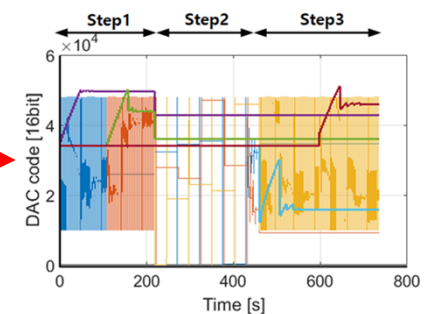


750um Substrate

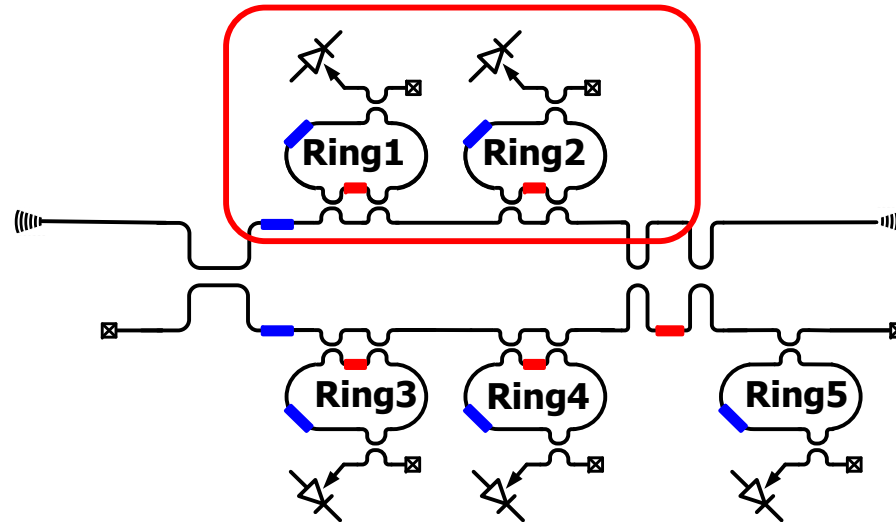


1042s reduction

75um Substrate

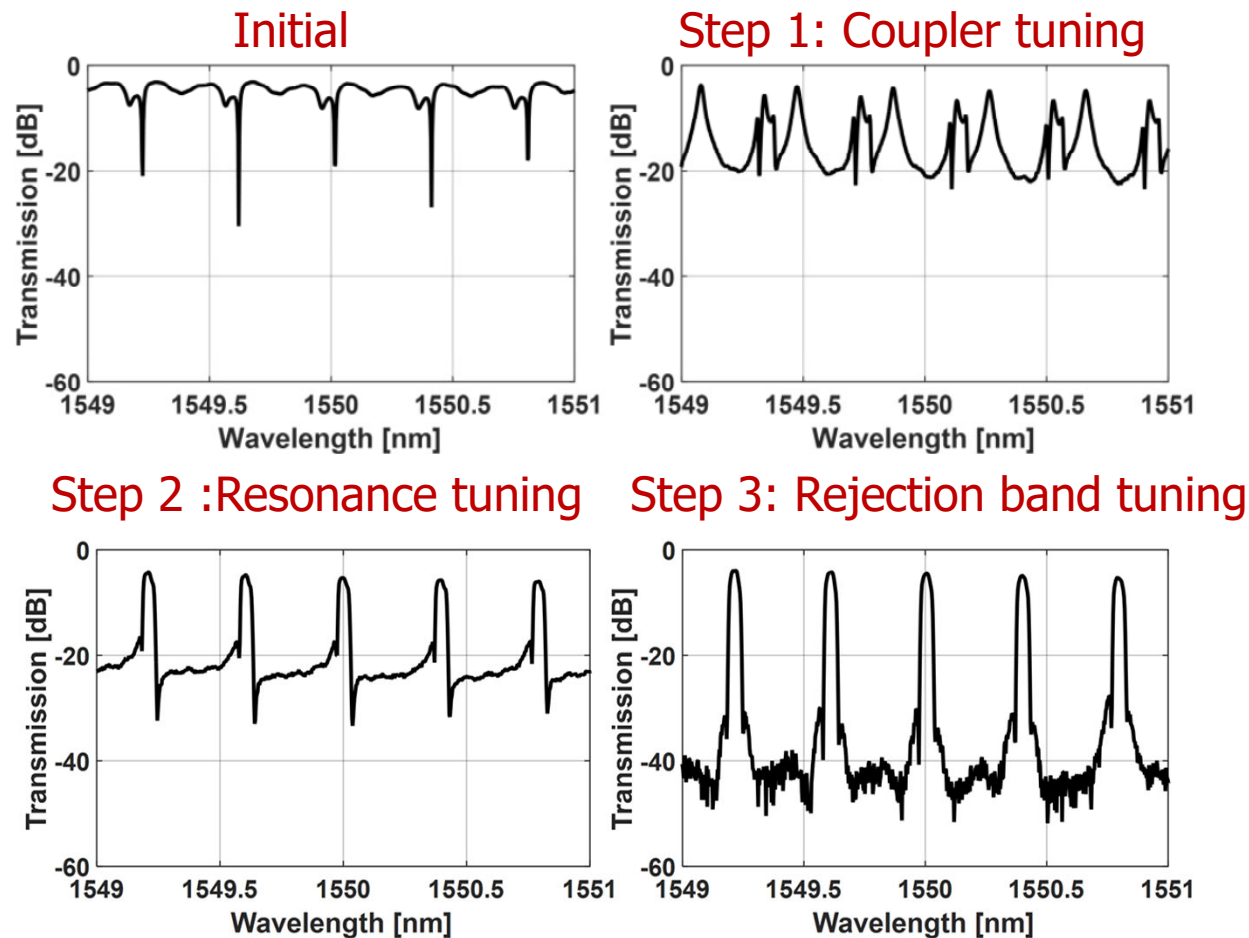


4th-order Filter Tuning

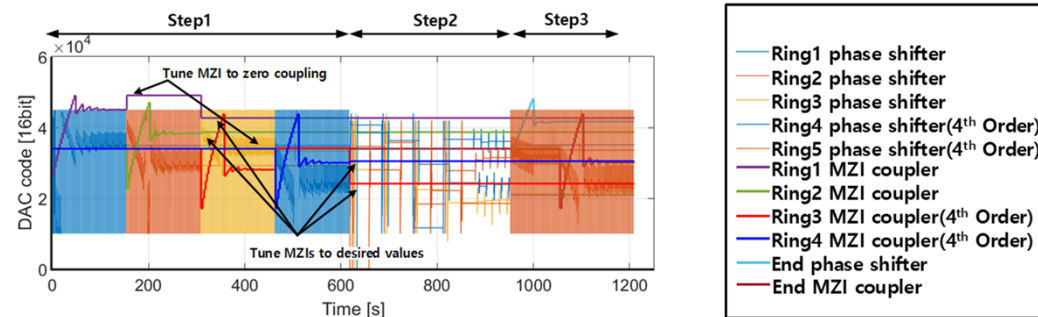


- What is different from the 2nd-order filter tuning?
 - Since two rings are cascaded, Ring2 and Ring4 monitor responses are influenced by the Ring1 and Ring3 response
 - Ring1/3 coupling factors are set to zero in order to tune Ring2/4 coupling
 - Increased number of iterations due to thermal crosstalk

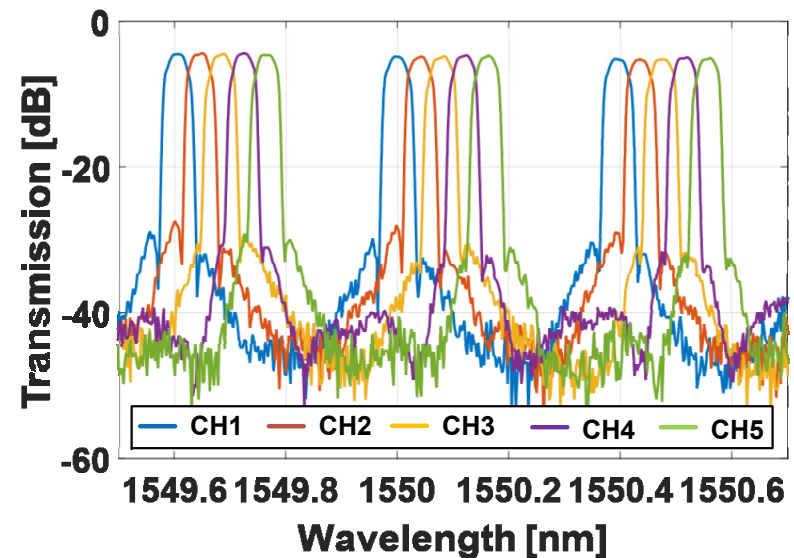
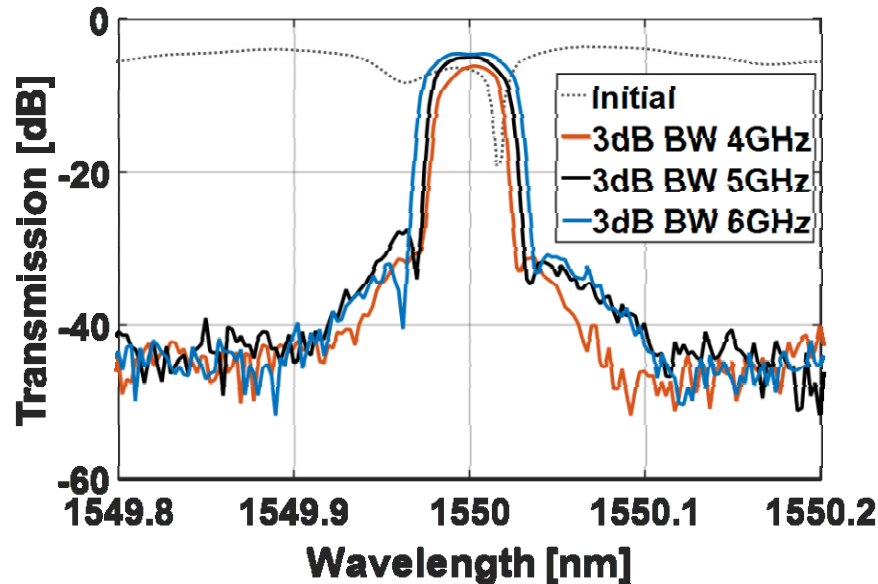
Measured 4th-order Filter Initial Calibration



w/ 75um Substrate



4th-order Filter Reconfiguration



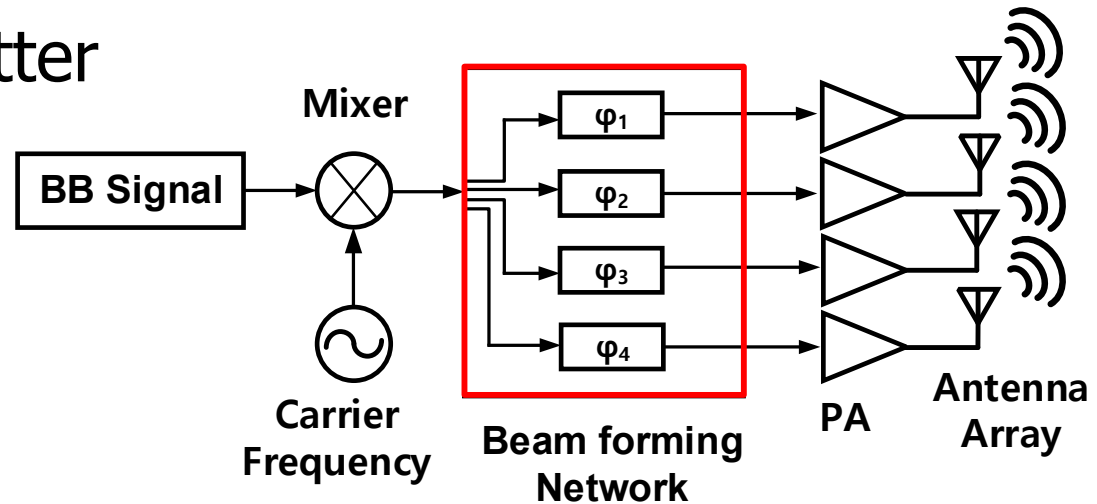
- Bandwidth reconfiguration to 4GHz and 6GHz 3dB BW
- Center frequency reconfiguration with 0.04nm spacing
 - 5 different calibrations performed with different laser center frequencies
 - After full calibration, center frequency switching only takes 300ms

Outline

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- Automatic Optical Beamforming Network Tuning
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True Time Delay Beamforming Network

- Phased array transmitter



- Beamforming network plays a critical role [3]
 - Beam focusing to the specific direction
 - Beam steering functionality
- Multi-GHz bandwidth, mm-wave frequency operation and high resolution beam angle tuning are required to fit into the 5G communication

Limitation of Electrical Beamforming Solution

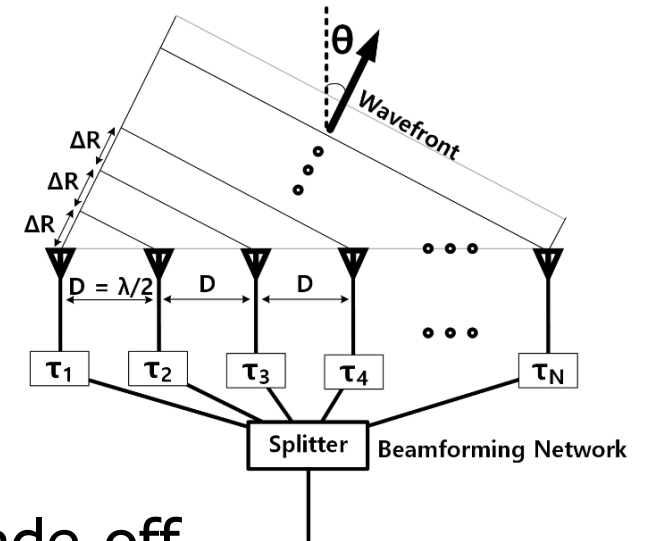
- Most of electrical beamforming network employ phase shifters in their RF path
 - Inherently narrow band
 - Beam radiation angle is dependent on the RF frequency (**Beam squint**)
 - Limited phase resolution (Discrete tuning)
 - Passive phase shifters are lossy
 - Active phase shifters are power hungry, linearity limited.
- Timed delay beamforming network
 - Squint free
 - Limited resolution (Discrete tuning)
 - Bulky and integration into CMOS is challenging

Beamforming Network Principle

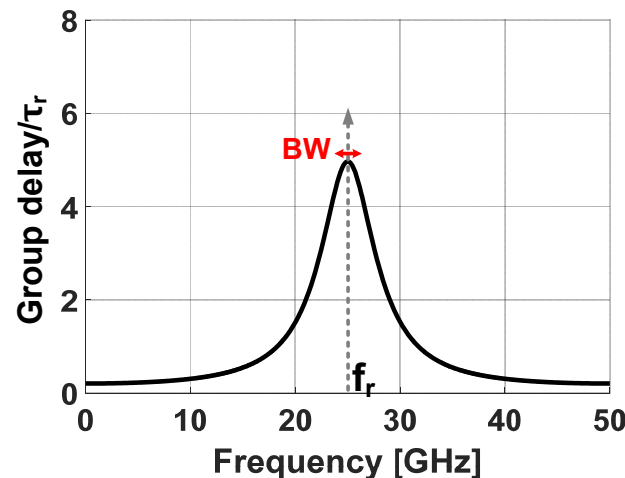
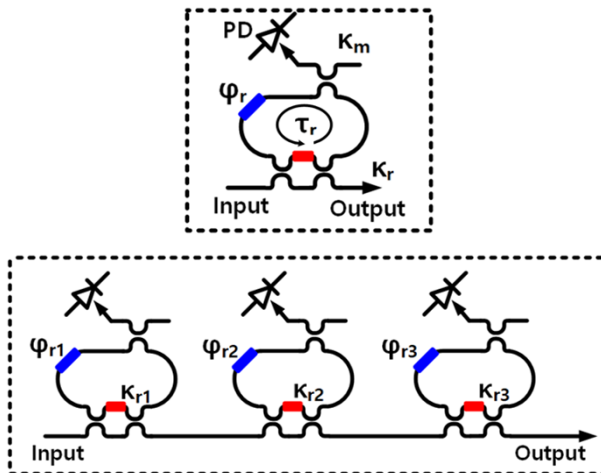
- Beamforming Network

Delay(τ) in Beamforming network
for radiating angle(θ) at the linear array

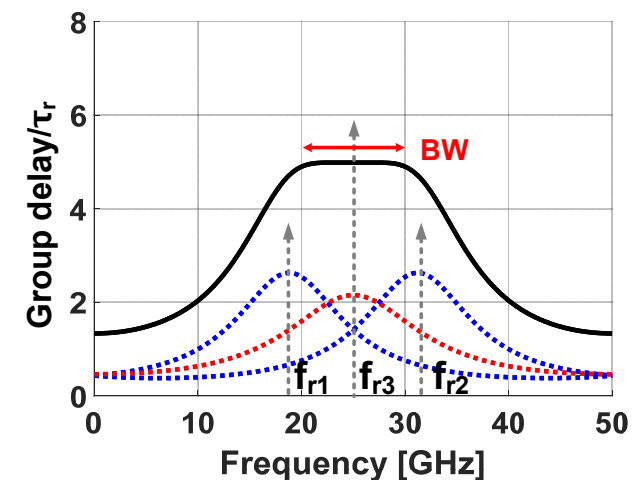
$$D = \frac{\lambda}{2} \quad \Delta R = D \cdot \sin\theta \quad \Delta\tau = D \cdot \cos\theta / c$$



- Single ORR has bandwidth-group delay trade-off
- Cascaded ORR can break the trade-off



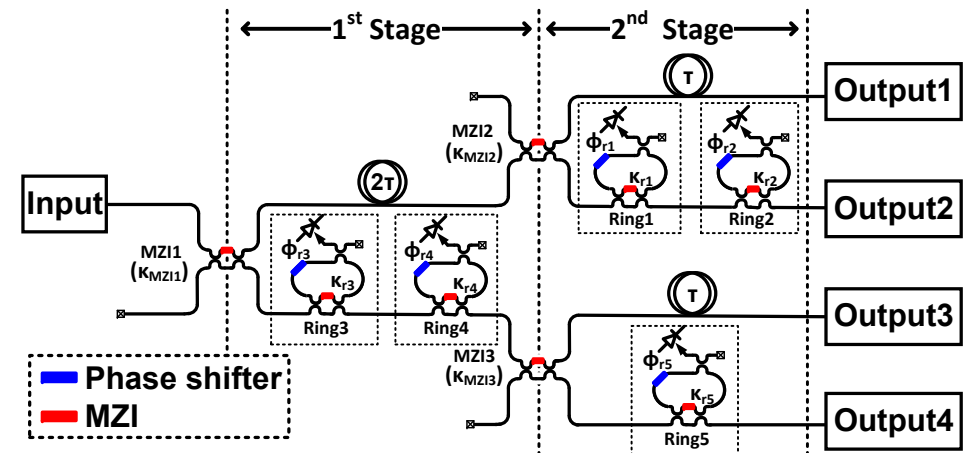
Single ORR



Cascaded ORR

Optical Beamforming Network (OBFN) Design

- Asymmetric binary tree structure
- Operating frequency: 30GHz
- Target Bandwidth: 2GHz
- Free spectral range: 50GHz
- Beam steering angle
 $-30^\circ \sim 30^\circ$ ($150^\circ \sim 210^\circ$)

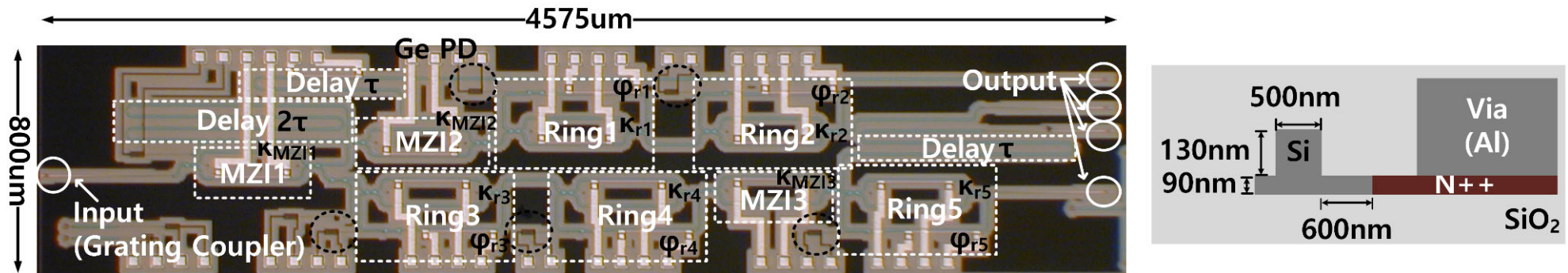


Group Delay Requirements

Radiating Angle (θ)	Output1	Output2	Output3	Output4
$\theta = 150^\circ$	0.0ps (0ps + 0ps)*	25.0ps (-24.4ps + 50.5ps)	50.0ps (-51ps + 101ps)	75.0ps (-76.ps + 151.5ps)
$\theta = 165^\circ$	0.0ps (0ps + 0ps)	29.0ps (-25.5ps + 54.5ps)	58.0ps (-51.0ps + 109.0ps)	87ps (-76.5ps + 163.5ps)
$\theta = 180^\circ$	0.0ps (0ps + 0ps)	33.3ps (-25.5ps + 58.8ps)	66.7ps (-51.0ps + 117.7ps)	100.0ps (-76.5ps + 176.5ps)
$\theta = 195^\circ$	0.0ps (0ps + 0ps)	37.6ps (-25.5ps + 63.1ps)	75.3ps (-51.0ps + 126.3ps)	112.9ps (-76.5ps + 189.4ps)
$\theta = 210^\circ$	0.0ps (0ps + 0ps)	41.6ps (-25.5ps + 67.2ps)	83.3ps (-51.0ps + 134.3ps)	124.9ps (-76.5ps + 201.4ps)

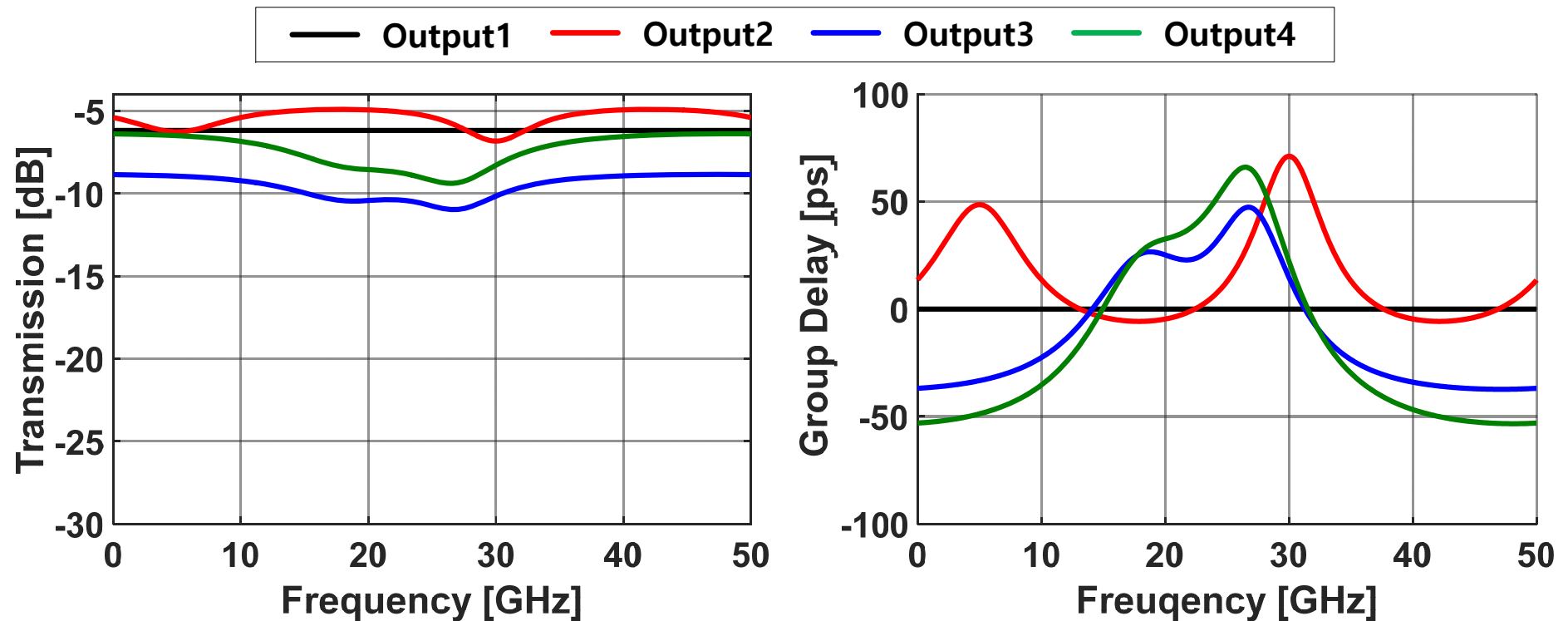
*Group delay (red: path delay, blue: ORR delay)

Silicon Photonic OBFN Prototype



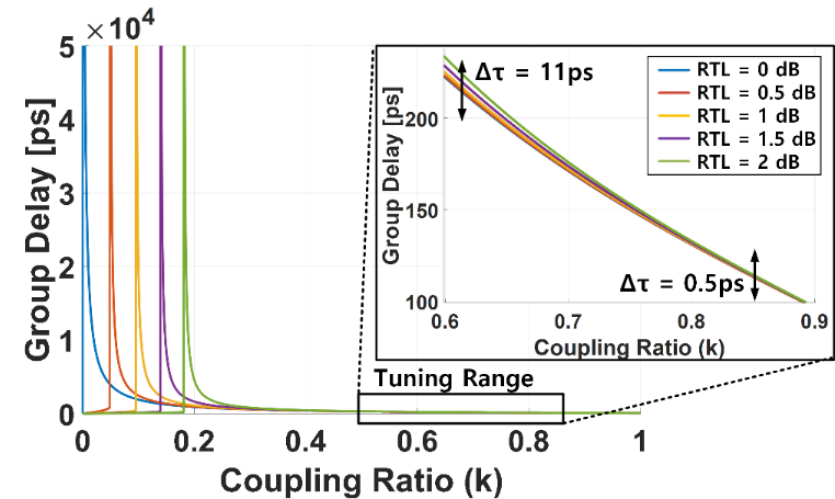
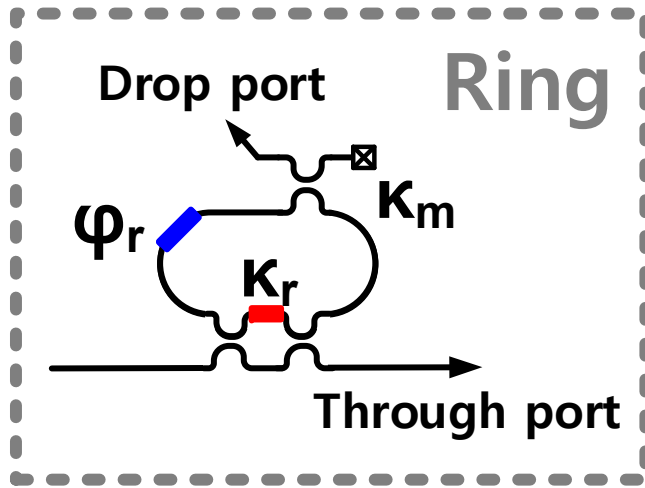
- Fabricated in IME SOI silicon photonics process
- Phase shifters are implemented with resistive heaters
- Rings are designed with 1554 μm circumference to provide an OBFN response with 50GHz free spectral range
- Delay lines (τ) are designed with 1554 μm length

Si-Photonic OBFN w/ Process Variations



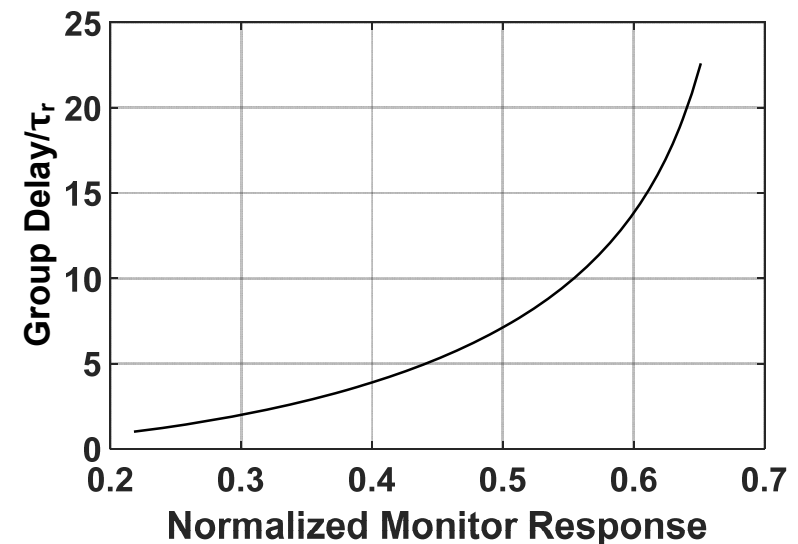
- Photonic devices are vulnerable to process and temperature variations
- Significant variation in 4-channel output power and group delay
- Need precise automatic calibration solution

ORR Group Delay Response

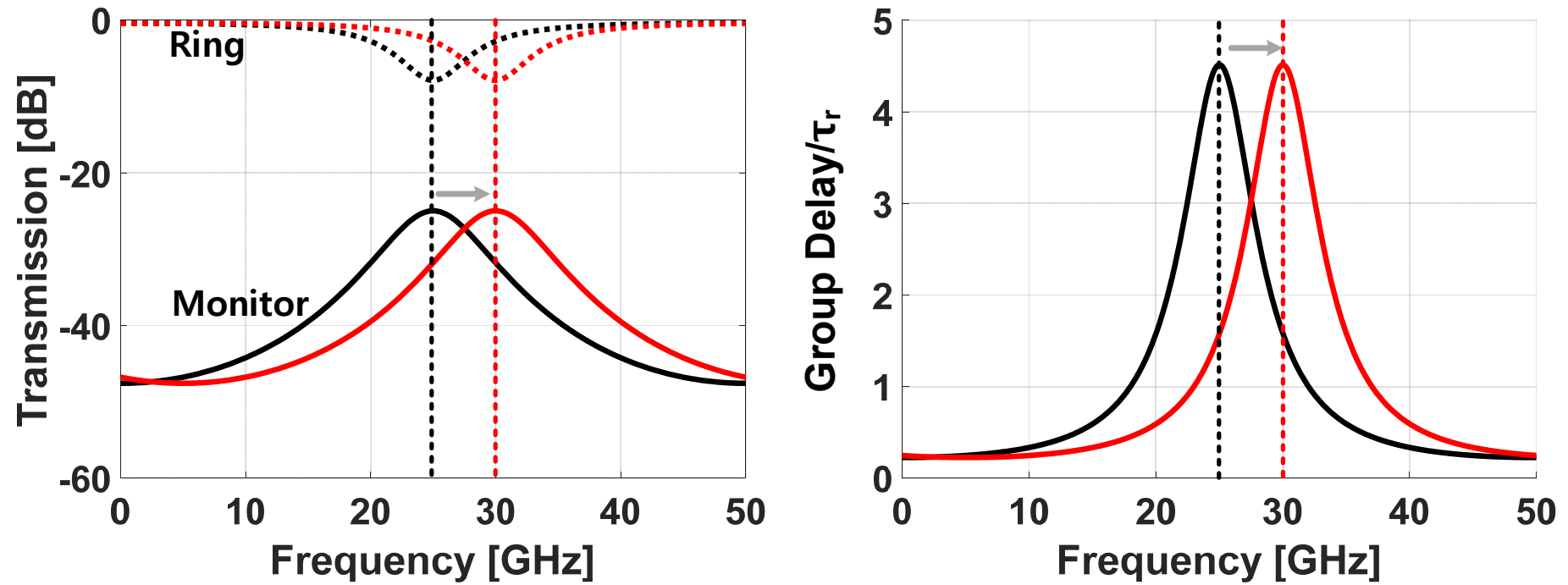


$$\tau_g(f) = \frac{\kappa_r \tau_r}{r(2 - \kappa_r) - (1 + r^2)\sqrt{1 - \kappa_r} \cos(2\pi f \tau_r + \phi_r)}$$

- ORR's coupling ratio and round trip loss (RTL) determines group delay peak value
- Group delay can be tuned based on monitor response

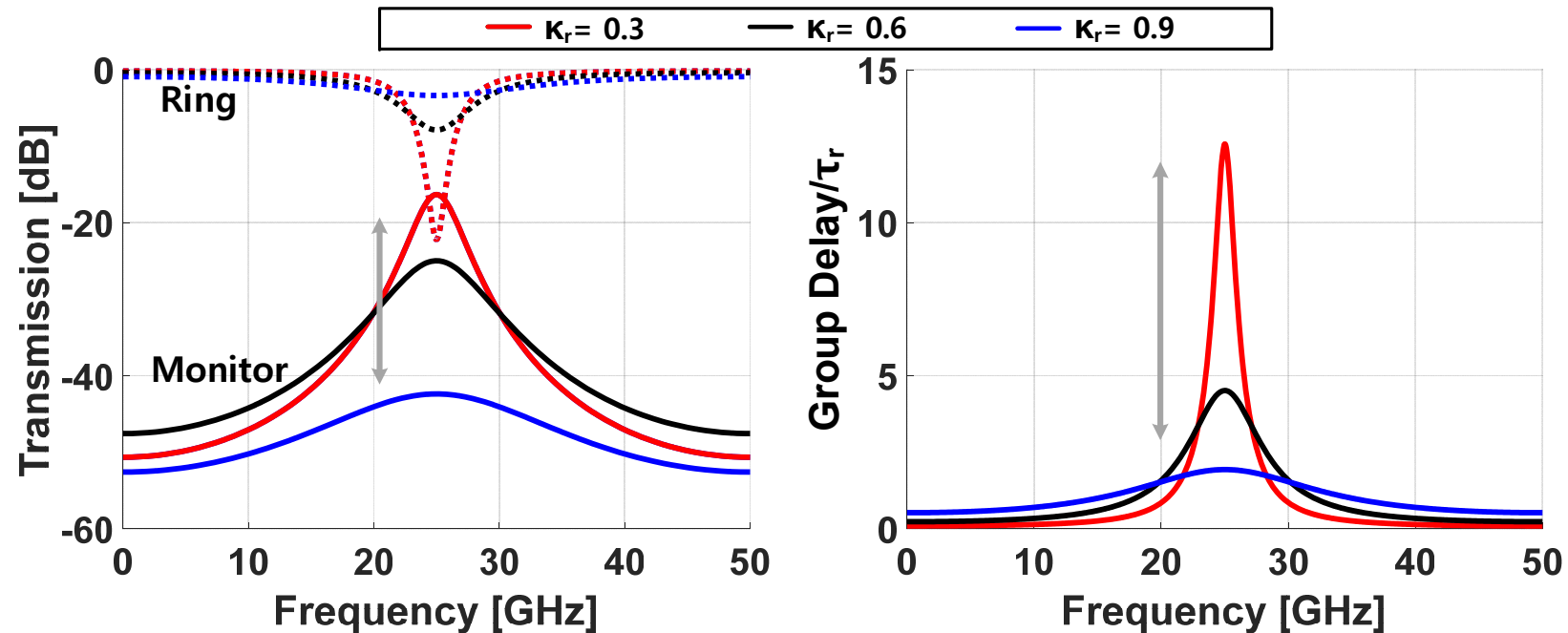


Monitor-Based Group Delay Tuning

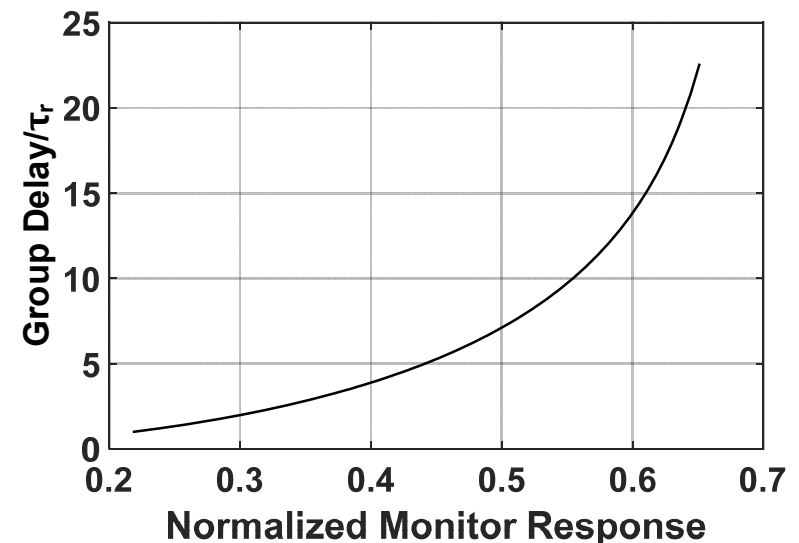


- Through-port group delay resonance is also aligned with through-port and drop-port magnitude resonance
- ORR group delay resonance can be tuned to the desired frequency through the ring resonance tuning procedure

Monitor-Based Group Delay Tuning

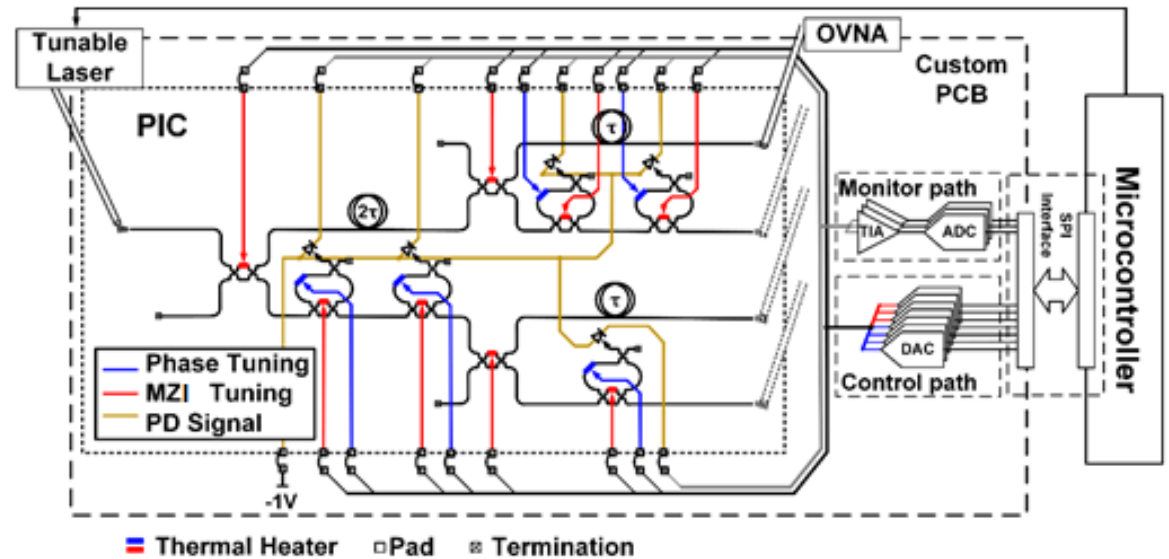


- Tuning of the coupling ratio can achieve the desired group delay response

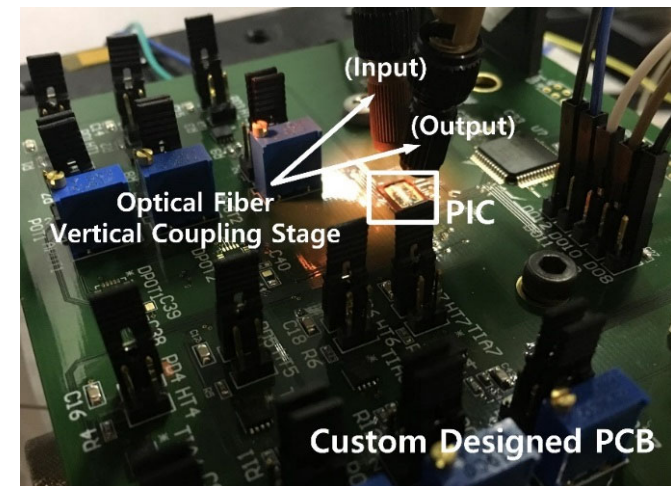


OBFN Tuning System and Procedure

- Automatic tuning system
 - Microcontroller
 - DAC: heater control
 - TIA, ADC: monitor

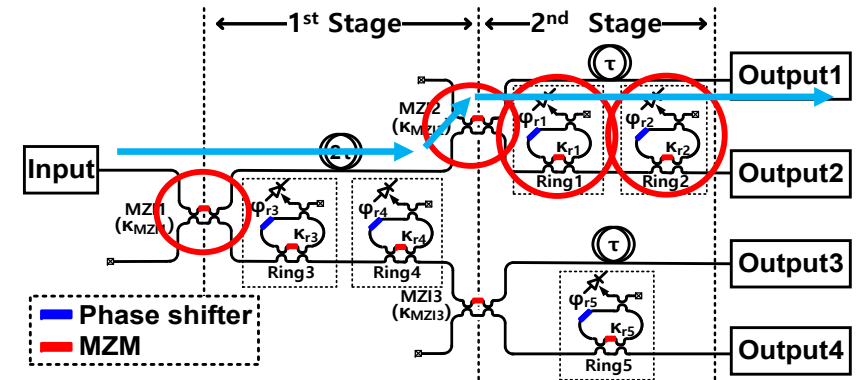


- Tuning procedure
 1. Output1 & 2 path tuning
 2. Output3 path tuning
 3. Output4 path tuning
 4. Resonance tuning

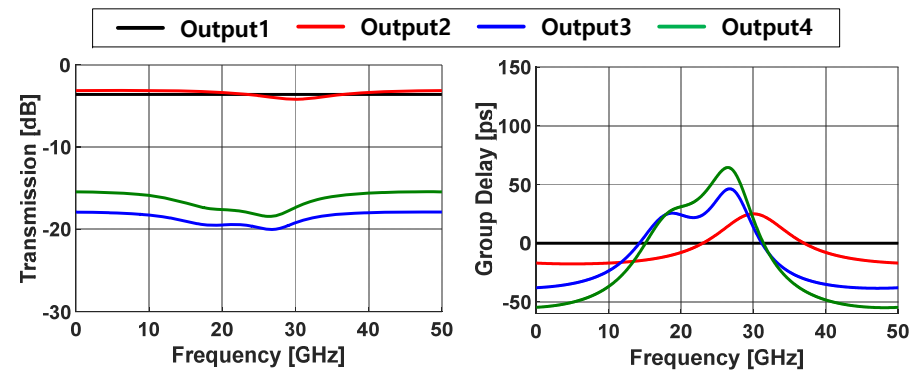


1. Output 1 & 2 Path Tuning

- Maximize monitor1 response while tuning MZI1&2
- Find critical coupling ratio of Ring1
- Find critical coupling ratio of Ring2
 - Set Ring1 to zero coupling
- Tune Ring1&2 MZI couplers to designed value
 - Using MZI characteristic and critical coupling ratio
- Tune MZI2 coupling to equalize output power
 - Using MZI characteristic and maximum coupling ratio
- Outputs 1 & 2 are now close to desired group delay at 30GHz and have near equal power, while Outputs 3 & 4 are still highly distorted

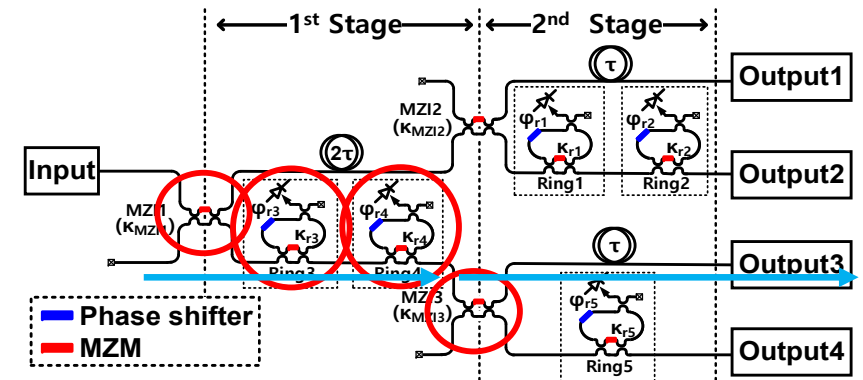


After Step1

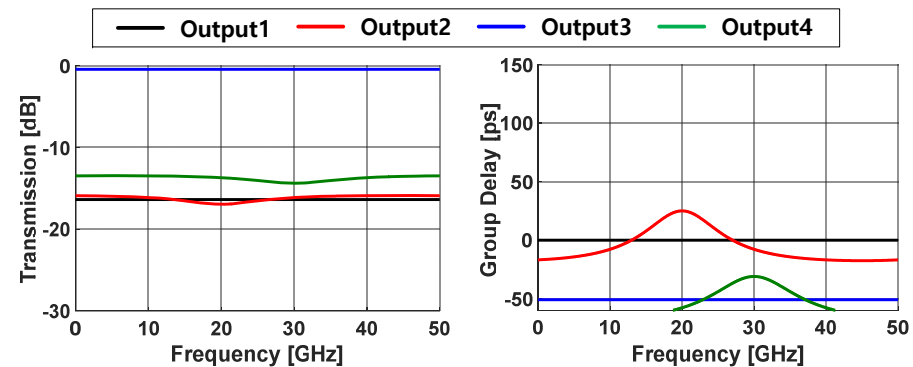


2. Output 3 Path Tuning

- Maximize monitor3 response while tuning MZI1
- Find critical coupling ratio of Ring3
- Find critical coupling ratio of Ring4
- Set Ring3 MZI coupler to zero coupling
- Tune Ring4 MZI coupler to zero coupling
- Using MZI characteristic and critical coupling ratio
- Tune to zero for Ring5 coupling tuning in Step3



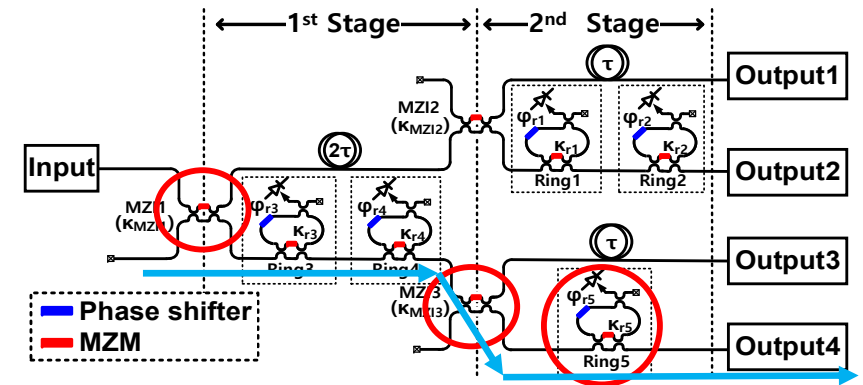
After Step2



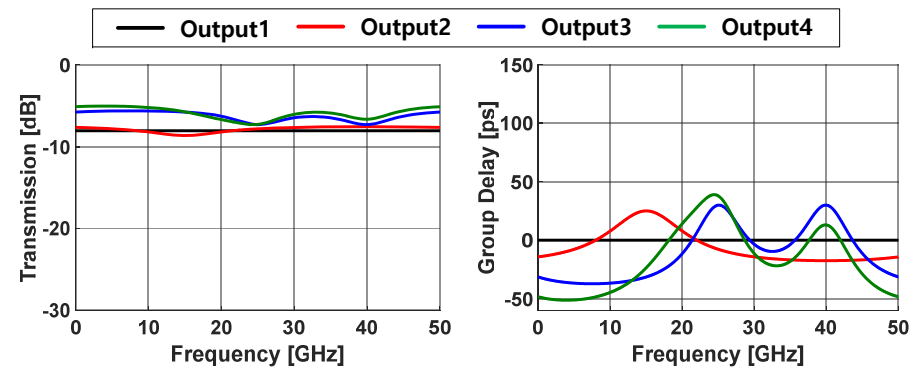
- After this intermediate step, the Output 3 group delay response is flat due to both Rings 3 & 4 coupling ratio being set to zero in preparation for the next step

3. Output 4 Path Tuning

- Maximize monitor5 response while tuning MZI3
- Find critical coupling ratio of Ring5
- Tune Ring3,4&5 MZI couplers to designed value
 - Using MZI characteristic and critical coupling ratio
- Tune MZI1&3 coupling to equalize output powers
 - Using MZI characteristic and maximum coupling ratio
- The 4 channel output powers are now closer, but group delays are still off due to the ring's resonance frequencies not being set

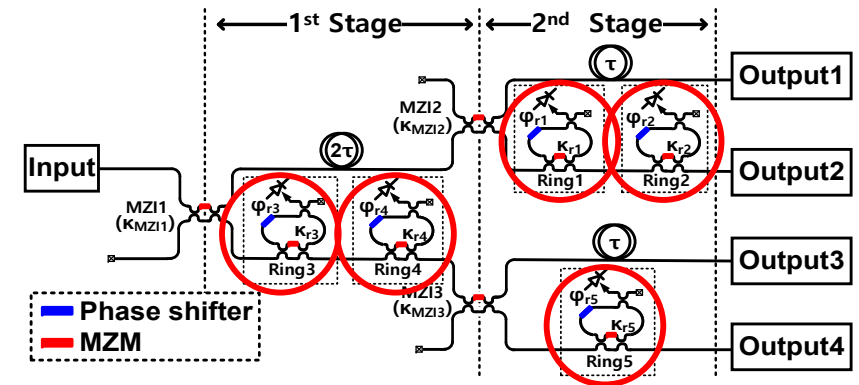


After Step3

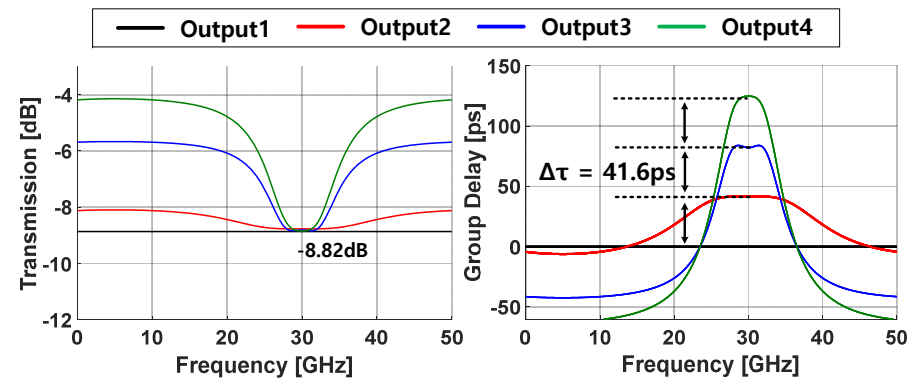


4. Resonance Tuning

- Resonance tune Ring1-5 to maximize the corresponding monitor reading
- Multiple iterative tuning to compensate thermal cross-talk
- Input laser frequencies are switched to center frequencies of each ring for targeted angle

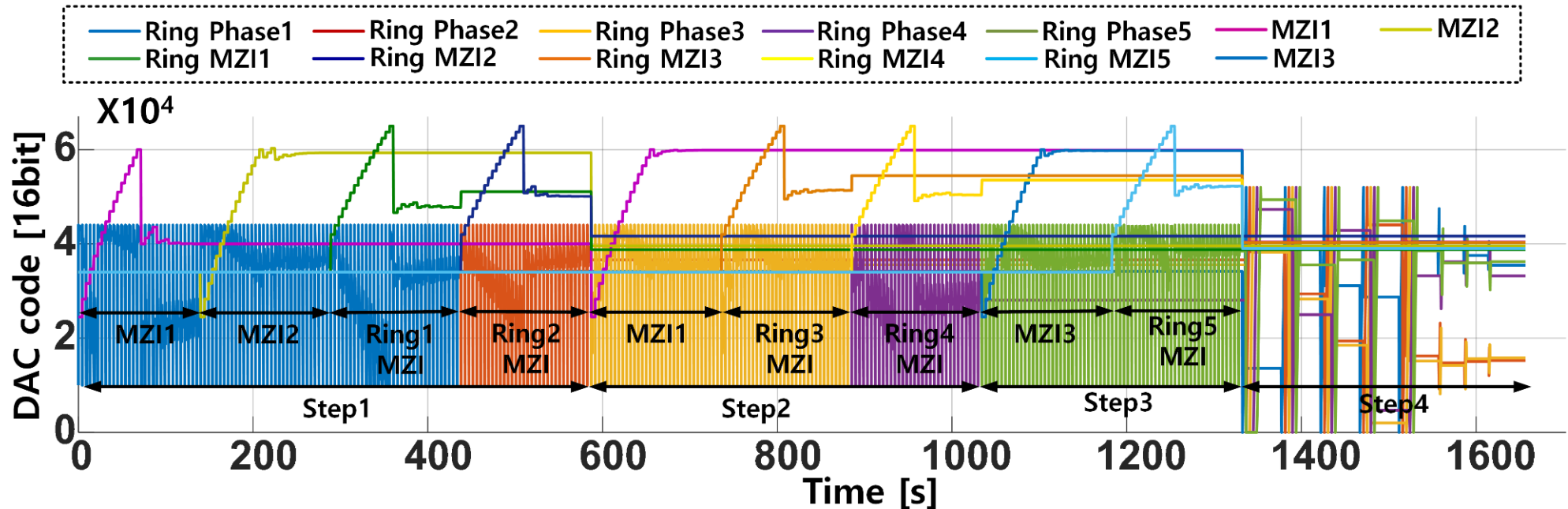


After Step4



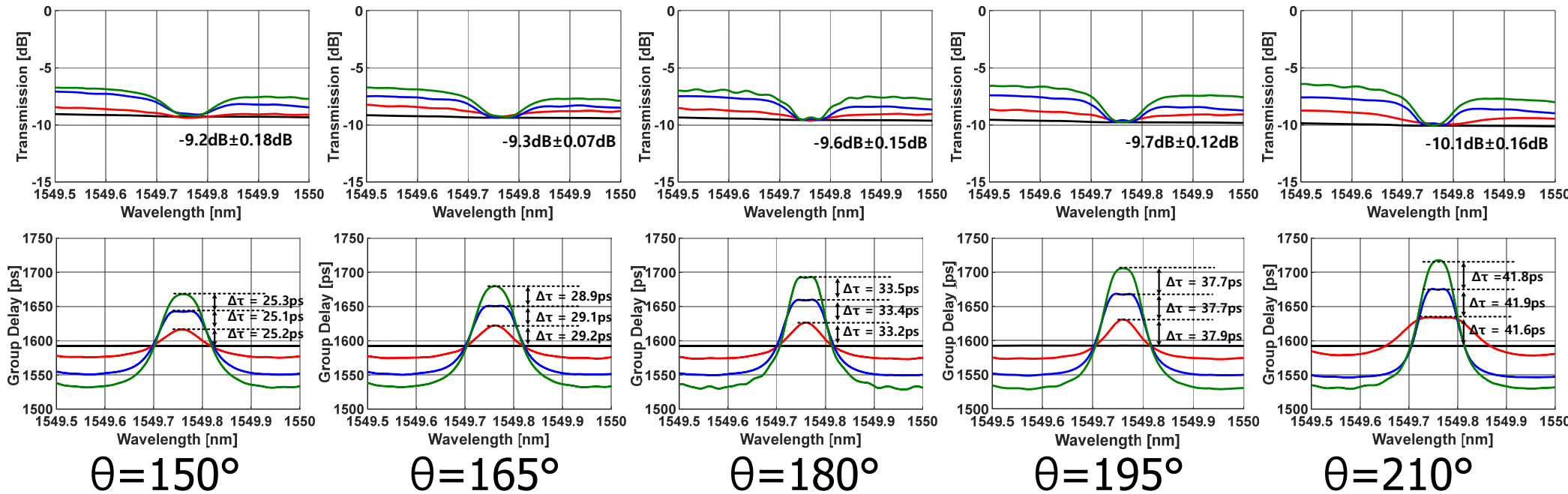
- The 4 channel outputs now have well-defined group delay responses and equalized power around the 30GHz center frequency

Measured Tuning Convergence



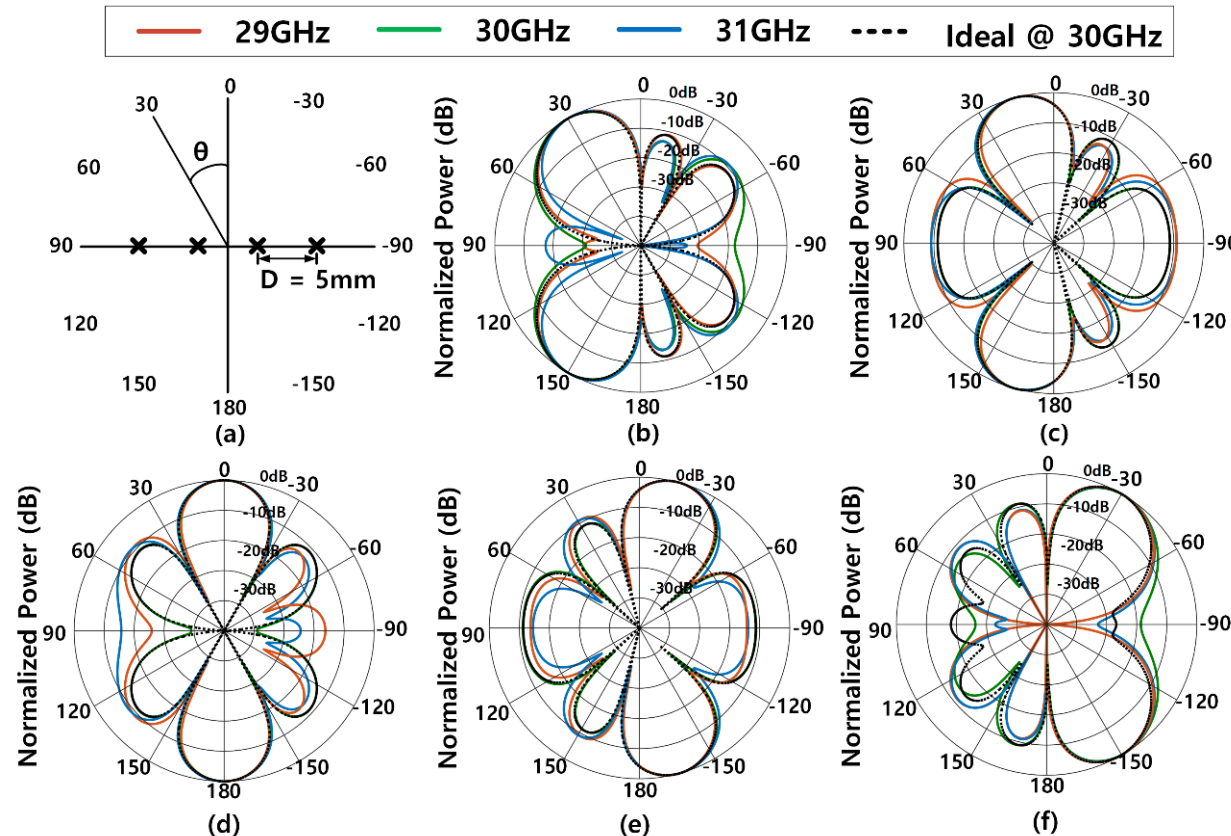
- Involves tuning 13 heaters
- While ORR MZI tuning, corresponding ring phase shifters are tuned in parallel
- Total tuning time 1617s

Measured OBFN Results



- Initial OBFN response is calibrated to have >2GHz bandwidth, centered at 30GHz relative to the 1550nm laser frequency
- Tuned result shows errors less than 0.3ps (OVNA resolution limit is 0.2ps)
- Each output showed power difference <0.2 dB mainly due to the grating coupler fabrication variations and alignment error

Beam Pattern Simulation Results



- 4-element linear antenna array with 5mm spacing beam patterns simulated based on measured OBFN responses
- Good directionality is achieved with main lobe showing at least 9.5dB larger gain than side lobes
- True time-delay operation of the ORRs allows for squint-free operation over 29-31GHz

Conclusion

- Automatic monitor-based calibration schemes developed for ORR-based photonic integrated circuits
- Severely degraded initial responses and reconfiguration demonstrated for 2nd/4th-order APF-based pole/zero filters and a 1X4 asymmetric binary tree OBFN
- Leveraging the proposed calibration schemes can allow for robust operation of these photonic structures in future wideband communication systems

References

Automatic Monitor-Based Tuning of APF-Based Pole/Zero Filters

- S. Cai, G. Choo, E. Zhian Tabasy, B. Wang, K. Entesari, and S. Palermo, "Adaptively-Tunable RF Photonic Filters," *IEEE International Midwest Symposium on Circuits and Systems*, Aug. 2015.
- S. Cai, G. Choo, B. Wang, K. Entesari, and S. Palermo, "Comprehensive Adaptive Tuning of Silicon RF Photonic Filters," *IEEE Texas Symposium on Wireless and Microwave Circuits and Systems*, Mar. 2016.
- G. Choo, S. Cai, B. Wang, C. Madsen, K. Entesari, and S. Palermo, "Automatic Monitor-Based Tuning of Reconfigurable Silicon Photonic 2nd-Order APF-Based Pole/Zero Filters," *OSA Conference on Lasers and Electro-Optics*, May 2017.
- G. Choo, C. Madsen, K. Entesari, and S. Palermo, "A Reconfigurable Silicon Photonic 4th-order Filter for Synthesizing Butterworth, Chebyshev, and Elliptic Responses," *IEEE Avionics and Vehicle Fiber-Optics and Photonic Conference*, Nov. 2017.
- G. Choo, S. Cai, B. Wang, C. Madsen, K. Entesari, and S. Palermo, "Automatic Monitor-Based Tuning of Reconfigurable Silicon Photonic APF-Based Pole/Zero Filters," *IEEE-OSA Journal of Lightwave Technology*, vol. 36, no. 10, pp. 1899-1911, May 15, 2018.

Automatic Monitor-Based Tuning of a 1X4 OBFN

- G. Choo, C. Madsen, S. Palermo, and K. Entesari, "Automatic Monitor-Based Tuning of RF Silicon Photonic True-Time-Delay Beamforming Networks," *IEEE International Microwave Symposium*, June 2018.
- G. Choo, C. Madsen, S. Palermo, and K. Entesari, "Automatic Monitor-Based Tuning of a Silicon Photonic 1X4 Asymmetric Binary Tree True-Time-Delay Beamforming Network," *IEEE-OSA Journal of Lightwave Technology*, vol. 36, no. 22, pp. 5263-5275, Nov. 15, 2018.